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# FORMULATION AND EVALUATION OF MOXONIDINE ORALLY DISINTEGRATING FILMS

**Abstract:**

Orally disintegrating films may accelerate the medication's onset of action, reduce dosage, and improve its efficacy and safety profile. It dissolves faster, is more stable, and lasts longer than other traditional dose forms. In order to create fast-dissolving films that dissolve in saliva rapidly and without the need for water, Moxonidine(MXD) was chosen for the sublingual route. The purpose of MXDoral disintegrating films was to increase patient compliance and bioavailability. The formulas were created with the intention of helping elderly, bedridden, and mentally ill people. MXD is used to measure the effect on the dissolution profile. The orally disintegrating films (ODF) were prepared using the solvent casting technique with water soluble polymers such as HPMC E3, HPMC E6, and Pullulan gum of various concentrations and PEG 400 as a plasticizer and an active ingredient. The medication dissipated rapidly, according to the results. This study suggests that an oral disintegrating film may be used to give MXD orally. ODF formulations dissolve smoothly and easily, releasing the entire medication within 20 to 30 minutes. Following disintegration, no remnant or particle matter was remaining. It was determined that the solvent casting method was effective in producing the quickly disintegrating MXD films. When there is no access to water, oral disintegrating films (ODF) have been effectively used with bedridden and psychotic patients as well as travelers. Prepared films demonstrated a higher rate of dissolution and a quicker start to action, which led to improved patient compliance, successful therapy, and increased use in the near future.

**Keywords:** Orally disintegrating films, Moxonidine, solvent casting technique, psychotic patients.

# INTRODUCTION

# CHAPTER-I

# INTRODUCTION

A lot of medication dosages are given as liquids, tablets, granules, and powders. Typically, a pill's design is intended to be chewed or swallowed whole to provide patients the exact quantity of medication. Under mild pressure, the pills—which come in tablet and capsule forms—are able to maintain their shapes. Nonetheless, some patients have trouble digesting or swallowing solid dosage forms, especially those who are young or elderly. For fear of choking, many elderly and pediatric patients refuse to eat these solid preparations [1].

Orally disintegrating tablets and Oro dissolving films are two of the fast-dissolving medication delivery technologies that have been developed as alternatives to traditional dose forms in order to help these individuals.

**Introduction to orally disintegrating system:**

Because oral medicine delivery has the highest compliance rate, especially among pediatric and geriatric patients, it is considered the most practical, affordable, and safest drug delivery method. Any medication delivery system's ultimate purpose is to successfully deliver the drug to the body. Among the different dosage forms, the oral disintegrating dosage form is the most popular commercial product [2].

Because it is the easiest to ingest, the oral cavity is the best location for the delivery of an oral disintegrating dosage form. Analgesics, neuroleptics, cardiovascular agents, antiallergic medications, and erectile dysfunction drugs are among the good drug possibilities for such a system. When applied on the tongue, a dose form like these dissolves rapidly, releasing the medication that dissolves in saliva. This is superior to a traditional dosage form in terms of absorption and quick beginning of action [3].

The oral route of drug administration remains the most essential mode of drug administration for systemic action, despite tremendous advancements in drug delivery systems. The oral route gained a lot of appeal since it was simple to administer and because it was conventional wisdom that the medicine would be absorbed just as well if taken orally as if it were food.

The traditional tablet appears to be the most widely used pharmaceutical dosage form due to its ease of transportation and relatively low manufacturing cost. However, a number of typical conditions that contribute to poor patient compliance include hand tremors, dysphagia in elderly patients, underdeveloped muscular and neurological systems in infants and recalcitrant people, and swallowing difficulties. An alternate oral dose type for treating the issue is the oral mucosal medication delivery system. Products that dissolve in the mouth, such tablets and films, may be more convenient and palatable for patients [4,5].

**Over view of oral cavity:**

Fast-dissolving oral delivery methods are solid dosage forms that, when placed in the mouth without being chewed or drank, dissolve or disintegrate in one minute. As the saliva descends into the stomach, pre-gastric absorption from the mouth pharynx and esophagus increases the therapeutic impact of the medicine after it has disintegrated in the mouth. In these circumstances, the drug's bioavailability is noticeably higher than that of traditional tablet dose forms [6].

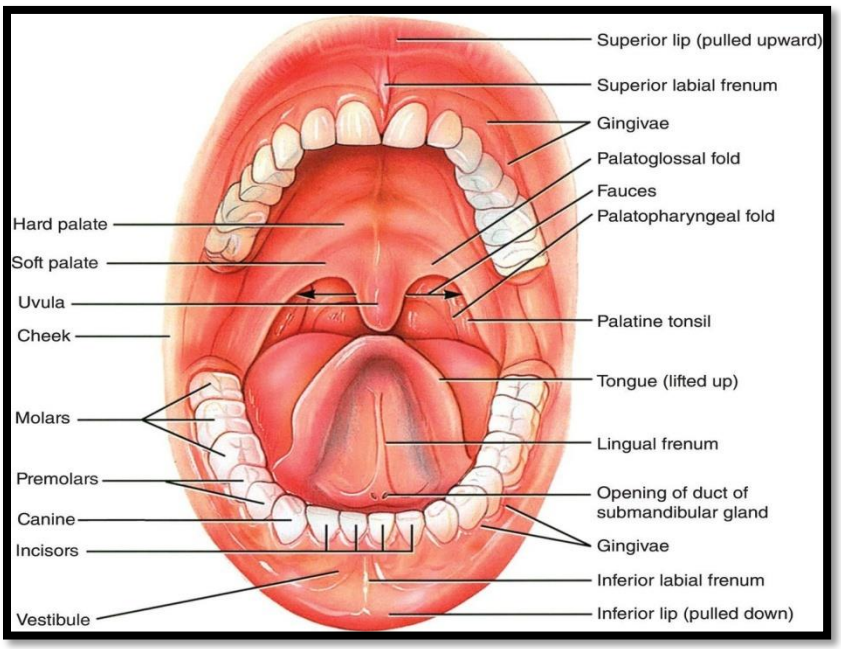


Figure 1: Diagram representing different parts of oral cavity

The stratified squamous epithelium that covers the oral cavity's surface is essentially kept apart from the lamina propria and submucosa, the underlying tissues, by an undulating basement

membrane [7]. It is noteworthy that the oral mucosa has a higher permeability than the epidermis, but a lower permeability than the intestines [8,9]. Additionally, it has been noted that the oral mucosa has a permeability that is 4–4000 times higher than the skin [10, 11]. Therefore, molecules with poor skin penetration can be absorbed very well through oral administration. Nevertheless, the main cause of the oral mucosa's permeability barrier is intercellular material that comes from the so-called "membrane coating granules" that are located at the topmost 200-micron layer [12, 13].

The oral cavity's epithelia are likewise made of mucus, an intercellular ground substance mostly composed of proteins and carbohydrates.

It concentrates defensive molecules like secretory immunoglobulins, keeps the oral cavity adequately lubricated, and lessens the ability of germs to adhere. Muco adhesive phenomena are caused by sialic acid residues and sulfhydryl groups found in negatively charged mucin [14]. The oral mucosa's barrier qualities are influenced by saliva and salivary mucin [15]. The lips, buccal mucosa, and the soft tissues of the mouth and throat contain small salivary glands [16]. The parotids through salivary ducts around the upper teeth, submandibular (tongue areas), and sublingual ducts make up the big salivary glands, which are composed of lobules of cells that release saliva. The total turnover rate of the main and small salivary glands' output, or 1-2 ml/min, is the flow rate under normal physiological conditions [17].

One of two ways that drugs can reach the oral cavity and be absorbed is either the paracellular channel, which travels between cells, or the transcellular route, which enters the cell by crossing the cell membrane. Because the sublingual mucosa is comparatively more permeable, absorption is accelerated and bioavailability is enhanced [18]. Compared to the sublingual mucosa, the oral mucosa is the preferable location for systemic transmucosal drug administration. Furthermore, the oral mucosa is more suitable for retentive systems utilized for oral transmucosal drug delivery since it is a relatively immobile mucosa and easily accessible. The main drawback of the buccal distribution method is that dose forms are difficult to remember for extended periods of time due to the oral cavity's continual release of saliva. Another drawback of buccal administration techniques is accidental ingestion of dose forms and salivary scavenging. The maximum time for buccal delivery is reported to be 4-6 hours [19].

Despite these difficulties, the buccal route continues to be the method of choice for delivering active pharmaceutical ingredients (API), which are very susceptible to gastrointestinal

tract degradation [20, 21]. Formulation development for youngsters has proven to be a difficult undertaking. The most important factor determining adherence to treatment plans, among other things, is the palatability of pediatric oral drug formulations [22, 23].

Younger children typically prefer liquid formulations because they are easier to administer, even if older children and teenagers generally tolerate solid dosage forms to ingest. Oral Disintegrating Dosage Forms (ODTs and ODFs) [24] are the result of pharmaceutical research with an emphasis on swallowing and ease of administration.

**Introduction to oral dissolving film:**

These traditional fast-dispersing or dissolving pills' main flaw is their solid physical structure. In certain populations, there is still a fear of swallowing, chewing, or choking on such solid-shaped items. Additionally, it is challenging to carry, store, handle, and administer wafer-like, porous, low-pressure molded tablets to patients—especially the elderly and children—due to their fragility/friability, which is caused by different manufacturing processes and necessitates the use of special, costly packaging to protect the dosage forms [25].

The development of Thin Oro Dissolving Film Technology has addressed the shortcomings of traditional fast dispersion or dissolving tablet formulations.

The film increases the risk of choking and the fear of choking, is simple to make, easy to handle and administer, and has handy packaging. It also lessens the taste that is unpleasant.

These thin polymer films are also known as melt-in-mouth dosage forms (MDF), mouth dissolving films (ODF), quick dissolving films (QDF), rapidly dissolving films (RDF), and oral dissolving films (ODF) [26].

Breath refreshing products have been the main market for rapidly dissolving films (RDF). RDFs, however, have also lately been made available for medicinal items in the US and Europe. When a film or strip is placed on the tongue, a water-soluble polymer inside the film or strip dissolves. Pfizer (New York) created the first oral strip as a mouth freshener ("Listerine" pocket packs). The first therapeutic oral thin-film product containing benzocaine was "Chlor-aseptic Relief Strips" (sold by Prestige Brands, Irvington, NY) for the treatment of sore throat pain [27,28].

The typical composition of an oral thin film is hydrophilic polymer, which can range in thickness from 50 to 200 nm, embedded with a medication that rapidly and completely melts and dissolves in saliva within the oral cavity, releasing the drug for absorption through the oral mucosa. A

portion of the medication will travel the length of the GI system and be absorbed along with the saliva [29].

**Advantages and limitations of orally disintegrating films [30, 31]:**

One or more of the following benefits may have served as motivation for the development of fast-dissolving intraoral dose formulations.

***Patient compliance:*** To make it simple for patients to ingest the dosage form whenever and wherever they choose so that it can quickly dissolve or disintegrate in the mouth to facilitate systemic absorption.

***Patient convenience:*** The purpose of developing these dosage forms to be taken without water is to improve patient convenience in terms of transporting and administering them, particularly when traveling.

***Rapid absorption and onset of action***: to cause quick absorption and a quicker start to therapeutic activity, mostly from dose forms that dissolve and disintegrate quickly, most likely as a result of quick absorption, dissolution, and disintegration. That is, antianginal treatment.

***Avoidance of first-pass effect:*** To increase bioavailability as a result of at least partially passing via the buccal mucosa, which partially avoids first-pass metabolism. As a result, bioavailability is raised.

***Elimination of water/improved stability:*** to give a substitute for liquid dosage forms (such as syrups, solutions, and dispersions) and so increase the drug's chemical and physical stability.

***Product life-cycle management:*** To increase the lifespan of a product by differentiating it from competitors and offering substitute formulations for traditional oral medications.

Despite their benefits, a lot of ODFs have restrictions on how much medication can be included in each unit dose. Additionally, because of the nature of ODFs, specific packaging is required, which could raise the price. It is not possible to deliver medications that irritate the mucosa or have an offensive odor or that are unstable at the buccal pH.

- The unique qualities of a film-based medication administration system:
- The film is thin and elegant, comes in a variety of sizes and forms, is non-obstructive, has mucoadhesion, disintegrates quickly, dissolves quickly, and releases quickly.

**Ideal characteristics of orally disintegrating films [32, 33]:**

**Fast disintegration:**

ODFs should dissolve in the mouth without the need for extra water; the patient's saliva supplies the disintegration fluid. It should be easy to swallow and have a comfortable mouth feel with the broken-up film. Although disintegration as quickly as possible is preferable, "fast disintegration" typically refers to films disintegrating in less than a minute.

**Taste of the medicament:**

ODFs often include the medication in a taste-masked form because the majority of medications are not appetizing. Drug taste masking becomes essential to patient compliance when delivery systems crumble or disintegrate in the patient's mouth, releasing the active chemicals that come into touch with the taste buds.

**Drug properties:**

The drug qualities should not have a substantial impact on the film property for the optimal ODF technology. The effectiveness of ODFs may be influenced by a variety of pharmacological features. For example, a drug's bulk density, hygroscopicity, particle size, crystal structure, and solubility can all significantly affect the final film qualities, like strength and breakage. Each medicine should have specific features that can be accommodated by the ODF technology.

**Film strength and Porosity:**

ODFs are intended to dissolve or disintegrate quickly, hence in order to guarantee speedy water absorption into the film, the porosity of the film is typically enhanced. Fast water absorption or wetting of the film and the dissolving of related particles into their component parts for quick dissolution are the films' primary characteristics. Excipients must be extremely wettable in order to do this, and the film structure must have a highly porous network.

**Moisture sensitivity:**

ODFs ought to be less susceptible to dampness. Since many highly water-soluble excipients are utilized in formulation to improve both fast dissolving and nice mouth feel, this problem can be particularly difficult to solve. Some of those extremely water-soluble excipients may even melt at high humidity levels due to their susceptibility to moisture. It is important to have a good packaging design or other method to shield ODFs from different environmental factors.

**Mouthfeel:**

Patients should be given a product that feels good in their mouths since mouth feel is important. Sometimes, even though the flavor is the only thing changed, a product's mouth feel perception can be enhanced by specific flavors, giving the impression that it is less gritty. Effervescence can be added to a product to help with disintegration and to improve mouth feel by lowering its "dryness."

**Taste masking:**

The appeal of ODF is what makes it successful. If an ODF product doesn't taste nice, it matters more than if its disintegration time differs by a few seconds. Patients regarded disintegration times ranging from 5 to 30 seconds as rapid, and goods falling within this range were acceptable provided they had a satisfactory flavor. It didn't matter how long the product took to dissolve if it tasted awful. Therefore, patients and doctors will look for alternative ODF if the product does not taste well. As a result, a lot of pharmaceutical companies are now utilizing ODF technology to provide doctors and patients with a greater selection of pharmaceutical actives spanning numerous therapeutic areas. Furthermore, a variety of factors, such as direct-to-consumer advertising, websites that provide medical information, and the rise in over-the-counter medication availability, have made customers more informed and demanding.

Additionally, consumers today have more options than ever before. When a substance—such as food, drink, or medication—is introduced in the mouth cavity, taste is perceived. Taste buds, which are the taste receptor organs, transduce a signal that causes this experience. Extremely sensitive nerve endings in these taste buds generate and send electrical signals to the parts of the brain responsible for taste perception through the seventh, ninth, and tenth cranial nerves. For a variety of reasons, more than 50% of pharmaceutical products are taken orally; one of the major formulation issues with some medications is bad taste. For medical professionals, administering bitter medications orally with a sufficient degree of palatability is crucial, particularly when treating elderly and pediatric patients.

Therefore, one of the key tenets of product evaluation in oral pharmaceutical formulation is the reduction or elimination of bitterness. The palatability of oral medicinal formulations has improved as a result of tried-and-true techniques for reducing bitterness.

A number of methods, including polymeric coating schemes, cyclodextrin complexation, ion exchange resins, liposome-based salt creation, microencapsulation, and the use of excipients

such as flavors, sweeteners, and surfactants, have been developed to improve taste.

Table 1: Flavor selection

| Basic Taste | Recommended Flavor |
|---|---|
| Sweet | Vanilla, Bubblegum, Grape fruit |
| Salt | Butter scotch, maple, apricot, peach, vanilla, wintergreen mint |
| Sour | Lemon, lime, orange, cherry, grapefruit |
| Bitter | Liquorice, coffee, chocolate, mint, grapefruit, cherry, peach, raspberry, orange, lemon, lime |

**Challenges in formulating orally disintegrating films:**

- Fast breakdown of the film
- Steer clear of thickening.
- Solubility in water
- The quantity of medication
- Possess adequate mechanical strength
- Shielding against dampness
- Little to no leftovers in the mouth
- Suitable for flavor masking technologies
- Excellent packaging design

**Selection of ODF drug candidates:**

The pharmacokinetic characteristics of an oral drug formulation (ODF), such as its peak plasma concentration, maximum plasma concentration time, and plasma AUC vs time curve for an equivalent dose of an ODF compared to a conventional oral dosage form, will vary based on the extent of pregastric absorption of the ODF. As a result, the ODF and the traditional oral dosage form are not bioequivalent. The literature provides examples where the same medication dose in an ODF's pharmacokinetic profile and bioavailability are not bioequivalent to the drug in a traditional oral dosage form. When compared to the same dose delivered in a traditional oral

dosage form, the pharmacokinetic profiles of ODF formulations of selegiline, apomorphine, and buspirone exhibit notable differences.

The following factors are taken into consideration when choosing a drug for oral dissolving and pregastric absorption in an ODF:

- No aftertaste of bitterness
- A dosage of less than 20 milligrams
- Moderate to small molecular weight
- Excellent soluble in both water and saliva
- Mostly nonionized at the pH of the oral cavity
- Capacity to permeate and divide into the upper GIT epithelium (log P>1, or better yet, >2)
- Capacity to enter the oral mucosa
- Not broken down by mucosal enzymes

The criteria of selection of a Dosage Form for an ODF include:

- Minimal cost per unit.
- All excipients approved by pharmaceuticals
- no need for a penetration enhancer
- a broad range of medication loading or dosages possible
- Low variability
- good bioavailability
- non-irritating
- On the other hand, several traits could make a medication inappropriate for administration as an ODF:
- brief half-life and repeated dosage
- Extremely bitter or unsatisfactory taste due to inability to perform flavour masking.
- Needs regulated or prolonged release. Composition of the system:

A mouth disintegrating film refers to a slender layer containing an active constituent, with dimensions ranging from 5 to 20 cm square. The fast breakdown in either saliva or water is achieved through the utilization of a distinctive matrix consisting of water-soluble polymers. Medications can be combined to provide a single dosage of 15 mg. The mechanical properties of

the films are significantly influenced by formulation parameters, such as reducing the glass transition temperature. In a typical composition, drugs constitute approximately 1-25% of the total, while fillers, colors, and flavors account for 0-40%, plasticizers for 0-20%, and water-soluble polymers for 40-50%.

**Drugs:**

A range of API, such as antihistamines (chlorpheniramine maleate), antiulcer (omeprazole), anti-asthmatics (salbutamol sulphate), antitussives, expectorants, antihistamines, and NSAIDs (paracetamol, meloxicam, valdecoxib), may be delivered via oral fast dissolving film technologies. However, the dosage form's size is restricted, making it challenging to insert high-dosage molecules into the films. The maximum amount of API that can be added to the film is 5–30 mg. The insoluble API is evenly distributed throughout the film. Depending on the final release profile, APIs can also be introduced in the form of milled, micronized, nanocrystals, or particles.

**Water soluble polymers:**

Film formers consist of polymers that are soluble in water. The utilization of film making polymers in soluble films has attracted significant attention in the fields of medicine and nutraceuticals. The rapid disintegration, favorable tactile sensation, and mechanical properties of the films are facilitated by the presence of water-soluble polymers. The deceleration of polymer disintegration is achieved by increasing the molecular weight of the polymer film basis. Some examples of water-soluble polymers commonly employed as film formers include HPMC E-3 and K-3, Methyl cellulose A-3, 6, & 15, Pullulan, CMC, PVP K-90, Pectin, Gelatin, Sodium Alginate, HPC, PVA, Maltodextrins, and EUDRAGIT-RD10. Polymerized rosin (36) is a novel polymer that exhibits the ability to create films.

The mechanical properties of films are significantly influenced by formulation considerations, namely the use of plasticizers. Plasticizers have been found to improve the mechanical properties of films, including tensile strength and elongation. Variations in the concentration of these qualities may have an impact on them. Polyethylene glycols, glycerol, and di-butylphthalide are among the plasticizers commonly utilized in many applications37.

In the context of pharmaceutical dosage forms or food products, the sweetening agent serves as the primary constituent responsible for dissolving or disintegrating within the oral cavity. Both artificial and natural sweeteners are included in order to enhance the palatability of

the formulation. Sweetening agents are commonly utilized at a concentration ranging from 3% to 6%w/w38,39, either as standalone agents or in conjunction with other substances.

Surfactants are utilized as agents for solubilization, wetting, or dispersion in order to rapidly degrade films and promptly release active constituents. Some often used chemicals include benzoyl chloride, sodium lauryl sulfate, tweens, and benzalkonium chloride. Poloxamer 407 is an essential surfactant that functions as a facilitator of solubilization, wetting, and dispersion.

The selection of flavor is dependent on the specific medication that will be incorporated into the mixture. The identification of an oral disintegrating/dissolving formulation by an individual is contingent upon two key factors: the persistence of the formulation's aftertaste for a minimum duration of ten minutes, and the initial perception of flavor quality within the initial seconds following consumption. The quantity of flavor required to mask a taste is contingent upon the specific type and intensity of the flavor. In formulation 40, it is recommended to incorporate flavors at a maximum weight percentage of 10%.

**Colour:** A wide range of colors is offered, including custom Pantone-matched hues, EU hues, FD&C hues, and natural hues. Their concentration is limited to a maximum of 1%w/w. Incorporating saliva stimulating chemicals can enhance saliva production, hence facilitating the rapid dissolution and release of quick dissolving films.

Some examples of these agents include citric acid, tartaric acid, malic acid, ascorbic acid, and succinic acid41. Taste-masking strategies encompass ion-exchange-resin compounds, flavors, and sweeteners. Utilising encapsulated APIs for flavor masking is challenging due to the potential issues with homogeneity caused by larger particles.

Table 2: Type of additives used for preparation of oral dissolving film

| Plasticizers | Sweetening Agent | Flavoring Agent | Coloring Agent | Saliva Stimulating Agent | Surfactant |
|---|---|---|---|---|---|
| Acetyl triethyl citrate | Mannitol, Sorbitol | Lemon | Natural coloring agent | Citric acid | Poloxamer 407 |
| PEG | Xylitol, Polyols | Peppermint | Titanium oxide | Lactic acid | Benzalkonium chloride |
| Propylene glycol | Aspartame | Cinnamon | Silicon dioxide | Malic acid | benzethonium chloride |
| Sorbitol | Glycyrrhizin | Vanillin | Zinc oxide | Ascorbic acid | Tweens |
| Glycerin | Saccharin, Cyclamate | Menthol | | Tartaric acid | Spans |
| Citrate ester | Mannitol, Isomalt maltitol | Wintergreen | | Sodium lauryl sulfate | |
| Triacetin | Acesulfame potassium | Orange | | | |
| Triethyl citrate | Dextrose, Fructose | Clove | | | |

**Different manufacturing processes employed in the formulation ofODF:**

The ODF can be produced using one or more of the following procedures:

- Solid dispersion extrusion
- Hot-melt extrusion
- Solvent casting
- Rolling
- semi-solid casting

**Solvent casting:**

In the solvent casting method, the drug and extra additives are dispersed in a suitable solvent using a high shear processor. This process results in the formation of a homogeneous, transparent, and viscous solution. Polymers that are soluble in water are initially dissolved in water. Once the viscous mixture has been degassed to remove any air bubbles, it is then

transferred into a petri dish and left to dry.

Table 3: Specification condition required by using solvent casting method

| Specification condition required by using solvent casting method | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Mixing Condition** | | **Agitated Emulsification Device** | | **Vacuum Defoaming Device** | | **Coating Apparatus** | |
| Temp | 20-90°C | Flow rate | 80L/h | Flow rate | 80L/ h | Passage time | 2-8min |
| Agitating time | 40-120 min | Agitating time | 15 min | | | Drying temp | 50-130°C |
| Rotating speed | 1000-2000 rpm | Homogenizer Pressure | 15 min | | | Solution temp | 40-90°C |

Table 4: Advantages and disadvantages of solvent casting method

| Advantages | Disadvantages |
|---|---|
| Great uniformity of thickness | Polymer must be soluble in a volatile solvent or water |
| Great clarity then extrusion | Viscosity should be formed |
| More flexibility | |
| Better physical properties | |
| Finished film thickness is typically 12-100μm | |

**Hot-melt extrusion:**

In hot melt extrusion method firstly, the drug is mixed with carriers in solid form. Then the extruder having heater, melts the mixture. Finally, the melt is shaped in to films by the dies[44].

Table 5: Solving pharmaceutical challenges of hot melt extrusion method

| Problem | Solution By Hot Melt Extrusion |
|---|---|
| Poor bioavailability due to Poor API solubility | Enhance dissolution |
| Poor API stability during processing caused by hydrolysis | No hydrolytic stress |
| Poor taste of the API | Taste masked dosage form |
| Manufacturing of film | Prepared various type of film such as oral film, buccal film etc. |

Table 6: Advantages and disadvantages of hot melt extrusion method

| Advantages | Disadvantages |
|---|---|
| Improved bioavailability of poorly soluble compounds | Thermal process (drug/polymer stability) |
| During processing no required solvents and water | Flow properties of the polymer are necessary to processing |
| Cost effective process with reduced production time and reduced number of unit operations | Limited amount of available polymer |
| Sustained, modified and targeted release capability | Require high power input |
| Better content uniformity was obtained among granules of different size ranges | The melt technique is that the process cannot be applied to heat-sensitive materials due to the elevated temperatures involved |
| Homogeneous distribution of fine particle occurs | Lower melting point binder risks situations where melting or softening of the binder occurs during handling and storage of the agglomerates |
| Superior stability at varying pH and moisture levels | Higher melting point binders require high melting temperatures and can contribute to volatility problems especially for here labile materials |

**Solid dispersion extrusion:**

The solvent casting method involves the dispersion of the medicine and additional additives in a suitable solvent by the utilization of a high shear processor. The aforementioned procedure leads to the creation of a uniform, translucent, and thick solution. Polymers with water solubility are first dissolved in aqueous solutions. After the removal of air bubbles, the viscous mixture is subsequently put into a petri-plate and allowed to undergo the drying process.

**Semi-solid casting:**

The initial step in the semisolid casting process involves the preparation of a water-soluble and film-forming polymer solution. The resulting solution is combined with a solution containing an acid-insoluble polymer, such as CAP or CAB. Subsequently, an appropriate amount of plasticizer is introduced in order to generate a gel mass. The gel mass is ultimately shaped into films or ribbons by the utilization of heat-controlled drums. The thickness of the film ranges from 0.015 to 0.05 inches. The acid resistive polymer and film-forming polymeric should be combined in a ratio of 1:439.

**Rolling:**

A drug solution is rolled on a carrier using the rolling method. There should be a special rheological consideration for the suspension or solution. The primary solvents are alcohol and water mixtures. After being dried on the rollers, the film is cut into the required sizes and shapes[47].

**Technologies used:**

**Soluleaves:**

Through the utilization of this methodology, the film is fabricated in a manner that facilitates the liberation of the active chemicals upon exposure to saliva. This approach is particularly advantageous for both elderly and young folks who may encounter challenges in ingesting conventional medications. Because of their unique formulation, SOLULEAVES are able to adhere to mucosal membranes and progressively release the drug over a 15-minute period 48. Foam burst:

A new patent for foam-filled film capsules was awarded in September 2004 under the name FOAMBURST. During the filming process, gas is blasted into the picture, giving it a

honeycombed structure. The film's gaps can be filled with gas, left empty, or packed with different substances to deliver active medications or create particular taste-burst effects. The capsules dissolve quickly due to their delicate honeycomb construction, giving them a melt-in-your-mouth feel.

**XGel:**

The XGel movie the pharmaceutical business was now able to benefit from a change in product offers and manufacturing methods thanks to technology developed by Bio Progress. XGel film may improve the stability of the product. In addition to being created for non-ingestible applications including cosmetics, ostomy pouches, sanitary products, and healthcare devices, the films can also be printed or coloured during the manufacturing process for branding and tagging, which is a helpful method to boost product identification50,51.

**Wafer Tab:**

An exceptional, cutting-edge, and extremely stable edible film dosage form is Wafer Tab. Wafer Tabs are a type of drug delivery where pharmaceutical active ingredients are combined with an edible film strip. The active pharmaceutical ingredient dissolves quickly and releases when the strip comes into touch with saliva in the mouth. It is also possible to flavour the Wafer Tab film strip for even better taste concealing. The active component is incorporated into the fused body. The film is a perfect way to administer medications that need to be shaped and sized precisely because it can be made in a range of forms and sizes. release as well as for usage by individuals who experience swallowing difficulties52.

**Limitations of ODF Technology:**

Thin film disintegration happens quickly (less than 30 s), however its drug loading capacity is limited to less than 30 mg. Drug loading may be enhanced by thickening the film or by employing numerous layers, however increased thickness may hinder breakdown.

**Quick- Dis Technology:**

Lavi pharm Laboratories Inc. (Lavi pharm) has successfully developed an intraoral fast-dissolving medication delivery system that effectively addresses the existing market demands. The intraoral medicine administration system, referred to as QuickDisTM, is a patent technology developed by Lavi pharm. It involves the use of a thin and flexible film that exhibits rapid dissolution. The film is positioned either at the floor or top of the tongue. The active ingredient is rapidly released at the application site, allowing for either systemic or local

absorption.

The Quick-DisTM drug delivery system offers a range of packaging options, including unit-dose pouches and multiple-dose blister packages.

The disintegration time of the Quick-DisTM film, which has a thickness of 2mm, typically ranges from 5 to 10 seconds, indicating the point at which the film begins to fracture upon contact with water. The dissolving time for a 2 mm thick Quick DisTM film, which refers to the point of which 80% of the film is dissolved in water, is roughly 30 seconds. A typical Quick-DisTM pharmaceutical delivery system demonstrates a 50% release profile of an API within a time frame of 30 seconds, followed by a 95% release profile within a duration of 1 minute.

Table 7: Examples of products using thin-film technologies

| **Supplier** | **Product** | **Active Ingredient (dose strength)** |
|---|---|---|
| Novartis | Various products under Theraflu and Triaminic brands | Phenylephrine hydrogen chloride (HCI) (2.5 to 10 mg)<br>Dextromethorphan hydrogen bromide (5 to 20 mg)<br>Diphenhydramine HCI (12.5 to 25 mg) |
| Pfizer | Benadryl | Diphenhydramine HCI (12.5 to 25 mg) |
| Pfizer | Sudafed | Phenylephrine HCI (10 mg) |
| MedTech Products /Prestige/ Brands | Chlo aseptic | Benzocaine (3 mg) Menthol (2 mg) |

**General properties and release mechanisms involved in ODF:**

ODF is a slender, non-adhesive film characterized by its low moisture content, rendering it convenient to manipulate, package, and administer. Labelling is also considered to be desirable.

The typical thickness of a film ranges from 1 to 10 mm, while its surface area varies from 1 to 20 cm2.

The film's rapid hydration facilitates immediate softening when applied in the mouth, while its low dry tack simplifies handling and application. The mucoadhesive and wet tack properties of the system are designed to effectively secure the film in position at the designated application site.

In the case of the ODF by a thickness of 2mm, the typical disintegration period, which refers to the point at which the film begins to fracture upon contact with water, ranges from 5 to 10 seconds. The dissolution time for ODF with a thickness of 2 mm, which is the duration required for at least 80% of the tested film to dissolve in aqueous solutions, roughly 30 seconds. Upon dissolution and disintegration, the drug is liberated from the dosage form. As the thickness of the film increases, the durations of disintegration and dissolution also increase. By altering the composition of the film, one can additionally influence the rate at which it breaks down and dissolves.

**Laboratory experiments conducted in a controlled environment:**

The physical and mechanical properties of the ODF drug delivery system are mostly governed by the manufacturing process. These parameters are commonly evaluated by in vitro testing methods, including measurements of thickness dry-tack, tensile strength, percentage elongation, rip resistance, and Young's modulus. Additional performance criteria, such as wet tack, bending length, disintegration/dissolution time, are assessed as part of quality control testing. A rapid dissolution medicine delivery method commonly demonstrates a 50% release of API over a time frame of 30 seconds, followed by a 95% release profile within a duration of 1 minute25. The calculations performed for the RDFs encompassed the determination of tensile strength, elastic modulus, and % elongation53. Tensile strength is the highest stress given to break a film. It may be calculated by taking the mean of three measurements and the cross-sectional area of the broken film, using the following equation.

The elastic modulus is the ratio of the applied stress to the corresponding strain (force in N) in the region of approximately linear elastic distortion on the load displacement profile. It may be calculated using the following formula.

$$\text{Elastic modulus} = \frac{\text{Force at corresponding strain}}{\text{cross-sectional area of the film}} \times \frac{1}{\text{corresponding strain}}$$

The following formula was used to get the percentage elongation:

$$= \frac{\text{Increase in length}}{\text{Original length}} \times 100$$

Table 8: Examples of currently market products in the form of ODF

| Trade Name | Manufacturer |
|---|---|
| Eclipse flash strips | Wrigley's |
| Cool mint/fresh burst Listerine | Pfizer |
| pocket paks | Novartis |
| Neocitran® ThinStrips™ Cough | Mattel/Momentous Solution, LLC.1M |
| night time cough | Novartis |
| Health strips | Innozen |
| Thera Flu thin strips | Innozen |
| Triminic thin strips | Pfizer/ Johnson & Johnson |
| Supress cough herbal | Apothecus Pharmaceutical Corp. |
| SudafedPE1M | |
| Ora film | |

**HYPERTENSION:**

Although frequently asymptomatic, hypertension—especially essential or primary hypertension—is a major risk factor for stroke and, to a lesser extent, ischemic heart disease. Thus, a key component in lowering the risk of cardiovascular disease is controlling hypertension. There are published national and international management recommendations.

**Definition:**

The pressure of blood on artery walls is what is commonly referred to as arterial blood pressure. It is commonly measured indirectly with a suitably calibrated sphygmomanometer in the brachial artery slightly above the elbow, and the result is given in millimeter-hours (mmHg). There are two measurements taken: the diastolic blood pressure, which is obtained during

ventricular dilatation, and the systolic blood pressure, which is obtained during ventricular contraction of the heart. Hypertension is defined as a blood pressure level above which an intervention has been demonstrated to lower the associated cardiovascular risk. It is characterized by a reading that is higher than "normal." Establishing an absolute norm for blood pressure is problematic due to its susceptibility to various influences and its bell-shaped distributions curve in the general population. Normal adult blood pressure is typically defined as having a systolic pressure below 130 mmHg and a diastolic pressure below 85 mmHg, or a ratio of less than 130/85 mmHg. Nevertheless, recent studies suggest that the optimal blood pressure level might be lower than the aforementioned value in relation to cardiovascular risk. Based on the criteria established by the United States, the current definition of normal blood pressure is less than 120/80 mmHg. Conversely, European and UK guidelines regard this value as exceptional.

Blood pressure readings between 130 and 139/85 and 89 mmHg are considered high normal or fall into the prehypertension category. Although diastolic blood pressure was the only factor used to characterize hypertension in the past, systolic blood pressure is now also considered to be significant in assessing risk, and current guidelines place equal weight on the two. Over 140 mmHg systolic and/or 90 mmHg diastolic blood pressure is typically regarded as hypertension. The phrases "mild," "moderate," and "severe" hypertension are commonly used, although they may be deceptive because absolute cardiovascular risk plays a more significant role in determining whether therapy is necessary and is dependent on factors other than blood pressure. Therefore, the majority of guidelines employ the following grading scheme to categories hypertension:

Grades 1 through 3 are 140–159/90–99 mmHg, 160–179/100–109 mmHg, and ≥ 180 /≥ 110 mmHg, respectively. According to US criteria, a grade of 1 corresponds to stage 1 hypertension, while grades 2 and 3 are included in stage 2. When there is a difference between the systolic and diastolic pressure values, the larger value is utilized for classification. Depending on the severity of hypertension, blood pressure readings collected many times over a variable length of time should be used to classify patients and inform treatment decisions [54].

In certain situations, ambulatory blood pressure monitoring may be employed. Although suggestions have been made, readings from ambulatory monitoring are typically lower than those from conventional measurement, and it is still unclear what constitutes normal and

unhealthy values. Severe hypertension that progresses quickly in malignant or accelerated hypertension is linked to retinopathy and frequently renal damage. A systolic pressure of 140 mmHg or more and a diastolic pressure of less than 90 mmHg are indicative of isolated systolic hypertension, which is primarily seen in the elderly.

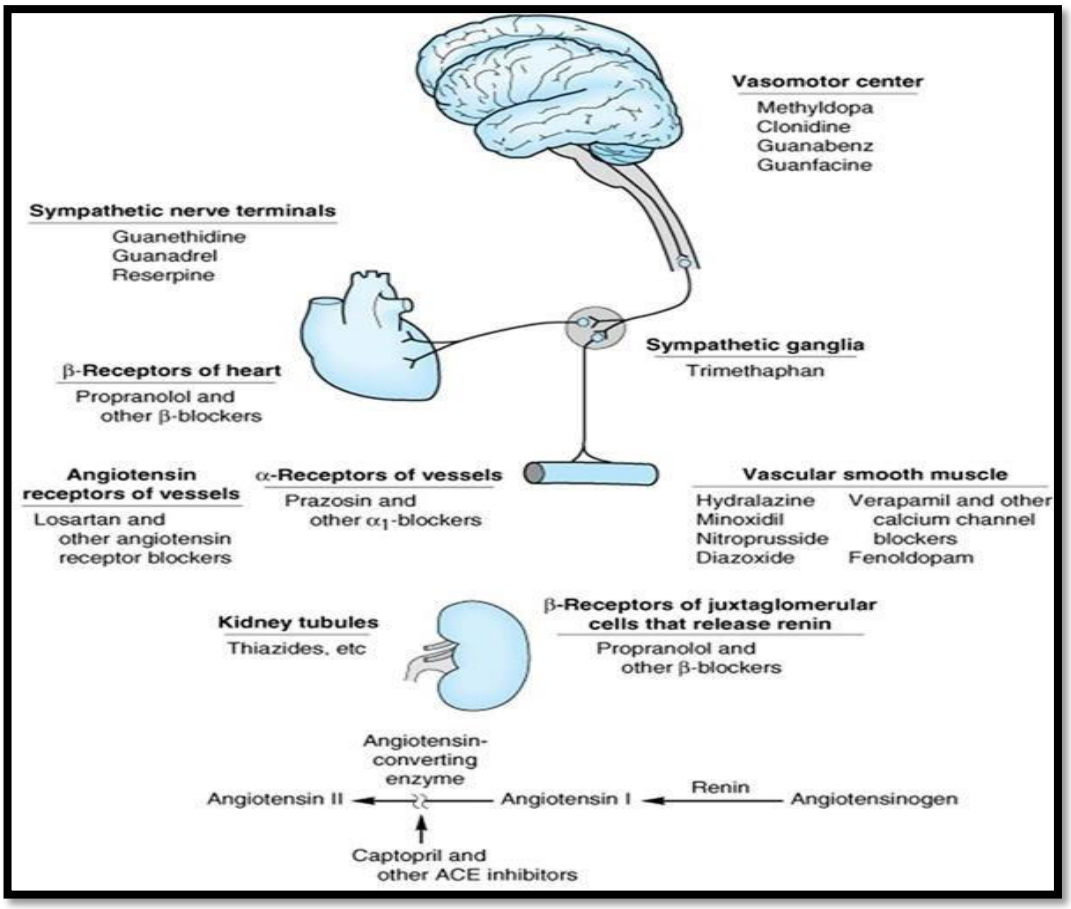


Figure 2: Sites of action of the major classes of antihypertensive drugs

Hypertension: Disease Information

Disorder Name: Hypertension

Type: Cardiovascular disorder

Hypertension has emerged as a significant global health issue, as per the European guidelines, wherein an office systolic blood pressure (SBP) of 140 mmHg and/or diastolic BP of 90 mmHg are considered acceptable [19]. Approximately 1.13 billion people worldwide are believed to be impacted, and it is projected that this figure will increase by 15% to 20% by 2025 [20, 21]. Hypertension, a significant global contributor to cardiovascular disease (CVD) and premature mortality, affects more than 60% of those aged 60 and above [22, 23]. The incidence

of hypertension increases as individuals age. Hypertension is commonly associated with obesity, diabetes mellitus, and chronic kidney disease (CKD), all of which contribute to an increased susceptibility to cardiovascular disease (CVD). Furthermore, hypertension has been found to independently induce harm to arteries and other vital organs [24, 25]. Hence, the primary goals of antihypertensive treatment are to mitigate end-organ damage, leading to a reduction in cardiovascular disease morbidity and mortality. Meta-analytic studies have demonstrated that a decrease of 10 mmHg in systolic blood pressure (SBP) or a reduction of 5 mmHg in diastolic blood pressure (DBP) is associated with a decrease in the relative risk of all major cardiovascular disease (CVD) events by roughly 20% [16]. This is an often observed cardiovascular ailment distinguished by a systolic blood pressure (BP) equal to or exceeding 150 mm Hg, as well as a diastolic BP equal to or exceeding 95 mm Hg.

The global prevalence of hypertension makes it a significant problem in the field of public health [27]. Hypertension is commonly acknowledged to be the underlying cause of about 7.5 million annual deaths, constituting approximately 12.8% of the total global mortality rate [28]. According to projections, the worldwide prevalence of hypertension is expected to reach 1.56 billion individuals by the year 2025 [29].

Regretfully, recent data from the US indicates that the percentage of persons with properly managed hypertension is declining (it is currently less than 45%), and that the death rate from hypertension-related CVD is rising in tandem. The trends are likely influenced by clinical factors such as insufficient treatments and treatment resistance, epidemiological factors such as population aging, rising prevalence of chronic kidney disease (CKD), diabetes mellitus, and obesity, and patient factors such as poor adherence. The COVID-19 pandemic has highlighted the importance of managing blood pressure, as persons having the disease been identified as hypertensive as a negative prognostic indicator [31]. The aforementioned findings have underscored the necessity for novel methodologies in the management of hypertension.

According to estimates, the prevalence of hypertension in the adult population as a whole was 26.4% in 2000. Furthermore, reports indicate that this prevalence rose in the USA, Mexico, and other countries, from 23.9% in 1994 to 29.0% in 2008; in Mexico, from 25.0% in 1993 to 43.2% in 2006; and in Canada, from 15.3% in 1995 to 24.5% in 2005 [34]. This prevalence makes it clear that hypertension is a major public health concern because its side effects—cardiovascular, cerebrovascular, and renal diseases—are the main contributors to morbidity and

death. In people with hypertension, lowering blood pressure either prevents or lessens these consequences.

In a small proportion of instances, hypertension can be attributed to a specific etiology. However, in the overwhelming majority of individuals (about 90%), the etiology remains unidentified. Consequently, the term "essential hypertension" is employed (37). The prevailing comprehension of essential hypertension posits that it is a multifaceted ailment resulting from the interplay of numerous hereditary, environmental, and behavioral factors. Due to the multifactorial nature of blood pressure homeostasis, any deviation in blood pressure, such as that resulting from a mutation, is expected to be counteracted by complementary actions, feedback loops, or adaptations to other regulatory mechanisms in an effort to restore blood pressure to its normal state. Essential hypertension occurs when there is a significant disruption in the equilibrium between the factors that contribute to elevated blood pressure and those that aim to restore it to a normal level, and the compensatory mechanisms are incapable of counteracting this disturbance. A century of studies on humans and animals in the fields of epidemiology, medicine, and physiology have yielded important new understandings of the connections between blood pressure, renal sodium processing, and dietary salt (sodium chloride). There is evidence linking a chronically high salt diet to the development of hypertension, wherein the kidneys cannot eliminate the quantity of salt taken unless blood pressure is elevated. Apart from the aforementioned primary causative factor, some adjuvant variables may also raise the risk of developing hypertension. The aforementioned factors encompass obesity, diabetes, the process of aging, emotional stress, a sedentary lifestyle, and inadequate potassium intake. As a result, some people get hypertension while others do not, and the likelihood that someone may develop hypertension is based on how much of the adjunctive factors for hypertension they currently have.

Figure 3: Blood cells flowing in the blood vessel

Elevated blood pressure, resulting from the circulation of blood within blood arteries and its resultant pressure on their walls (Figure 3), represents a notable risk factor for the development of coronary heart disease, stroke, and chronic heart disease. Elevated blood pressure is strongly correlated with an increased risk of stroke and coronary heart disease. In addition to coronary disorders and stroke, the implications of this condition encompass peripheral vascular disease, heart failure, renal impairment, retinal hemorrhage, and visual impairment.

Table 9: Systolic and diastolic blood pressure range

| Category | Systolic (in mm Hg) | Diastolic (in mm Hg) |
|---|---|---|
| Optimal | Less than 120 | Less than 80 |
| Normal | Less than 130 | Less than 85 |
| High Normal | 130-139 | 85-89 |
| Hypertension | | |
| Stage-I | 140-159 | 90-99 |
| Stage-II | 160-179 | 100-109 |
| Stage-III | 180 or higher | 100 or Higher |

# REVIEW OF LITERATURE

# CHAPTER-II

## REVIEW OF LITERATURE

**Nishi Thakur et.al** Drafted the "Overview." "A novel method for treating patients with quickly dissolving films." Rapid dissolving drug delivery systems are becoming more well-known and well accepted as innovative medication delivery methods. They seek to improve patient compliance while optimizing a drug's safety and effectiveness by forming it into a traditional oral dose form for administration. The film, which dissolves quickly, is applied on the tongue's surface or its floor. Upon application to the tongue, this film rapidly dissolves, so releasing the drug, which subsequently dissolves in saliva. As saliva descends into the gastric cavity, specific drugs are assimilated through the oral cavity, pharynx, and esophagus. Enhancing drug bioavailability in this particular scenario mitigates the risk of asphyxiation and provides a favorable oral sensation. A rapid dissolution medicine delivery approach is employed to overcome the challenges associated with ingesting pills, capsules, and similar substances. This review article provides a comprehensive overview of the advancements in the utilization of oral dosage forms, factors to be taken into account during formulation, methods for preparation and evaluation, commercially available products, and patented technologies for oral fast-disintegrating films.

**Arunarya et.al** were examined "Quick-dissolving oral films: A novel approach to medication administration and dosage form." Dissolvable oral thin films, or OTFs, originated as breath strips in the confection and oral care industries and have now developed into an innovative and extensively used method for distributing vitamins and personal hygiene items. Companies who have successfully created polymer coatings containing APIs for the purpose of delivering medicine through the skin have experienced advantages by transitioning their technology to organic drug delivery (ODF) forms.

**G. Kadhe and R.E. Arasan et.al** investigated in an effort to provide a description of the advancements in oral hypoglycemic agent drug delivery, especially the instant release forms of **MXD**.

**Shelen M colham et.al** In the collaborative Atorvastatin diabetes trial, participants with type 2 diabetes examined atorvastatin's primary preventive effect on cardiovascular disease.

**Jigishapatel et.al** examined dyslipidemia in people with diabetes. According to the study, stats

enhanced cardiovascular outcomes in those with diabetes mellitus. People with type 2 diabetes who did not have cardiovascular disease saw a substantial reduction in cardiovascular events at 3.9 years when taking 10 mg of atorvastatin daily.

**Salim Bastaki et.al** An investigation was conducted on diabetes mellitus, a metabolic disorder resulting from a defect in the activities or secretion of insulin. Insufficient production of insulin leads to the development of chronic hyperglycemia and disruptions in the metabolic processes of fat, protein, and carbs. As the condition deteriorates, there is a heightened likelihood of acquiring diabetes complications such as ulceration, neuropathy, retinopathy, and nephropathy due to tissue or vascular damage. Diabetes includes a wide range of various disorders.

**Wale kiran K et.al** were developed, "Formulation and development and in-vitro evaluation of sitagliptin phosphate monohydrate immediate release tablet." The goal of this study is to create a sitagliptin phosphate monohydrate instant release tablet formulation that is stable, enhanced in quality, and pharmaceutically similar. The sitagliptin phosphate monohydrate instant release tablets were made utilising the wet granulation process with varied concentrations (2–8%) of sodium starch glycolate, crospovidone, lactose, and microcrystalline cellulose as super disintegrants (S1–S9). Prepared formulations were tested for all relevant evaluations.

**Aggarwal Jyoti et al.,** Has conducted a comparative analysis of several polymers used in the manufacture of oral fast-dissolving films for alcohol and propylene glycol. The investigation was conducted with various polymers. Pullulan gum was used with various polymers, but only HPMC E15 and sodium alginate demonstrated the necessary film-forming ability. The disintegration time increased with higher polymer and plasticizer concentrations. According to the study's findings, pullulan films had the best ability to create films and were smooth, homogeneous, and clear. Additionally, the study found that Pullulan films had the shortest disintegration time. Out of all the film forming agents, Pullulan has the best film forming property.

**PrasannaDesu et al.,** Developed and assessed fast-dissolving films containing the antimigraine drug zolmitriptan. A film-forming substance called hydroxyl propyl methyl cellulose was employed. The films were created via the solvent evaporation process, and FTIR and UV analyses were used to characterize them. Propylene glycol was employed as a plasticizer in this study. After screening films for drug content, it was discovered that 98.62% of them could load drugs per 2 cm2. For the films, taste masking was also used. It was discovered that the folding

endurance, dissolving time, and dissolution profile were all satisfactory. The findings indicate that no interaction among Zolmitriptanand excipients. After the selected films were further assessed, it was determined that there was no bitterness in the film and that the Zolmitriptan HPMC rapid dissolving films broke down in just one minute.

**Sandeep D Jadhav et al.,** A study was conducted to find the optimal formulation of levocetirizine dihydrochloride through the examination of fast-dissolving oral films. These films were developed using numerical optimization and optimal design approaches. The independent variables in this study were sodium CMC and the film-forming component HPMC. Responses were obtained for drug release kinetics and release ranging from 45 to 990 seconds. A change from zero order to Hixson-Crowell based release was observed when the viscosity of the film decreases below a specific threshold. Among the various formulations, the optimal formulation exhibits a delivery time of 360 seconds for the whole administration of the medicine. The IR and DSC experiments indicated that there was no interaction among the drug and excipients. The process of drug release was facilitated by disintegration, and the impact of pH and agitation intensity was examined. The stability experiments demonstrated that the improved formulation exhibited stability. The effective development of a rapidly dissolvinglevocetirizine2HCl was achieved through the utilization of PEG-400, sodium CMC, and HPMC as significant excipients in the optimization phase. The measured independent variables exhibited a high degree of proximity to the anticipated values of the optimal formulation.

**Shelke PV et al.,** In the present study, sodium alginate was employed as the polymer for film formation in order to fabricate a film of Amlodipine Besylate that exhibits rapid dissolution. In order to reduce the disintegration time of formulations, sodium starch glycolate was used as a disintegrating component. The dependent variables in this study were the disintegration time and % of drug release, while the independent variables were the concentration of the polymer and disintegrant. The study employed a complete 32 factorial design. After six minutes, optimized batch disintegration demonstrated adequate drug release and was finished in the shortest amount of time. After six minutes, all of the formulations showed a 70–85% drug release.

**Ravneet Kaur et al., (2012)** The purpose of this study was to investigate the film-forming characteristics of several film formers used in oral film technology and to optimize plasticizer concentration for good folding endurance and tensile strength. Solvent casting was the procedure used to prepare the films. The prepared strips' ability to form films, appearance, disintegration

time, tensile strength, and folding durability were assessed. Polyethylene glycol (PEG 400) was utilized as the plasticizer in varying amounts (0.1, 0.2, and 0.3). The study found that HPMC E5 and HPMC K100 polymers had the least amount of disintegration time, the best film forming capacity, and an extremely translucent look. For good tensile strength and folding durability, HPMC E5 is the best film forming agent out of all the film forming polymers when combined with PEG400 at a 0.2 ml ratio.

The study conducted by **AjaykumarPatil et al. (2011)** focused on the development and evaluation of a rapid-dissolving oral film targeted at montelukast sodium. The film was fabricated by the solvent casting technique, employing PEG 400 as a plasticizer and HPMC as the structural material for the film. The film was supplemented with different amounts of super disintegrants, including crospovidone and microcrystalline cellulose. An evaluation was performed on the physicochemical characteristics of the fast-dissolving films. The compatibility of the drug with the formulation was confirmed using IR and DSC studies. The scanning electron microscopy technique was employed to elucidate the morphology of the films. The formulations containing 10% MCC and 4% crospovidone exhibited a maximum cumulative percentage drug release of 94.64% and 97.42%, respectively, for a duration of 30 minutes. Following a storage period of three months at a temperature of 40º20C and a relative humidity of 75±5%, there were no significant alterations seen in the physical properties, in vitro disintegration time, or drug content (4% cross povidone) of the formulation. The findings indicated that the combination of 4% crospovidone and 10% MCC, along with 4% HPMC as a film substrate, exhibited favorable performance in the production of rapidly dissolving monocost sodium films.

**Nidhi P. Sapkal et al., (2011)** The current study set out to create oral thin films of the mucolytic agent ambroxol hydrochloride and examine how formulation factors such as the concentration of the plasticizer, emulsifying agent, and film-forming polymer affected the in vitro dissolution tests and the product's physicochemical characteristics. The film former utilized was hydroxypropyl methylcellulose, the plasticizer was polyethylene glycol (PEG) 4000, and the emulsifying agent was Tween 80. These three factors were examined on two different levels, leading to the development and evaluation of formulations using the solvent casting process. HPMC had a crucial part in determining the film's properties. Folding endurance, tensile strength, moisture absorption percentage, disintegration duration, and in vitro dissolution rate were all greatly impacted. It was also discovered that PEG 4000 influences the characteristics of

films. It was discovered that the film with low concentrations of HPMC (15 mg/film, or 2x3 cm2), high concentrations of Tween 80 (5 mg/film, or 2x3 cm2), and PEG 4000 (8 mg/film, or 2x3 cm2) was ideal for film creation with desired physicochemical features, quicker disintegration, and optimal in vitro release.

**Yoshifumi Murata et al., (2010)** Investigated on films that dissolve quickly. These films were made from naturally occurring polysaccharides, like pullulan, and weren't heated, had their pH adjusted, or had any additional components added. Three distinct films containing either lidocaine, pilocarpine, or dexamethasone were made and used in this investigation. We looked into the release characteristics of the model pharmaceuticals from the movies. When there was no medication, the polysaccharide formed a circular film as a result of the casting process and the solvent's subsequent evaporation. Film formation was impacted by the type and concentration of medicines present. The concentration of the polysaccharide could be changed to control the film's thickness, and the 2% pullulan film's surface showed consistent unevenness. All polysaccharide-prepared films expanded easily in the dissolution media, liberated the integrated component, and then fell apart. Within fifteen minutes, all of the dexamethasone had been released from the films; however, the rate of release was marginally slower than that of lidocaine or pilocarpine. As a result, FDFs made from polysaccharides may be good options for medication dosage forms taken orally, as they should dissolve pharmaceuticals when in the mouth.

**Mahesh A et al., (2010)** Using cyclodextrins' capacity to mask taste, fast-disintegrating films of levocetirizine dihydrochloride have been produced. These films are useful for treating chronic urticaria and acute allergic rhinitis. Solvent casting was used to create the rapidly disintegrating films. The films included pre-gelatinized starch as a disintegrant, aspartame and sucrose as sweeteners, and water-soluble polymers like pullulan or Kollicoat® IR. Levocetirizine dihydrochloride was added to these films through the in-situ creation of a combination with beta-hydroxypropyl cyclodextrin. The optimized films were assessed using a human gustatory sensation test for weight variation, film thickness, folding endurance, tackiness, tensile strength, assay, content homogeneity, and in vitro and in vivo disintegration and dissolution as well as taste masking ability. The findings showed that complexing levocetirizine dihydrochloride with hydroxy propyl beta-cyclodextrin enhanced its organoleptic characteristics and allowed for the effective formulation of the complex into a fast-disintegrating film.

**M Koland et al., (2010)** An investigation was conducted on the fast-dissolving films of Ondansetron hydrochloride, a highly effective antiemetic medication. Ondansetron hydrochloride is a 5 HT3 antagonist with a short half-life of 5 h and a bioavailability in the oral cavity of 60% due to hepatic first pass metabolism. The present investigation was undertaken to develop and evaluate rapidly dissolves ondansetron HCl films designed for sublingual administration, with the aim of overcoming the aforementioned constraint. The films were fabricated by the solvent casting technique, employing a combination of polymers including PVA, PVP, and Carbopol 934P in different proportions. In addition, propylene glycol or PEG 400 was employed as plasticizers, while mannitol or sodium saccharin served as sweeteners. Based on the IR spectrum observations, there has been no interaction observed between the drug and polymer, as well as with other additives. Satisfactory findings were obtained when conducting tests to assess the homogeneity of weight, thickness, surface pH, folding endurance, homogeneity of drug content, swelling index, bioadhesive strength, and tensile strength. In addition, vitro drug release investigations were conducted using films in conjunction with USP dissolving equipment investigations on the release of drugs in vivo with USP dissolving device. Studies on drug permeability in vivo were conducted using a membrane model of pigs. Ex vivo studies showed 66-80% release within 7 minutes, while in vitro release tests showed 81-96% release within 7 minutes. Within 40 minutes, 66-77% of the drug had permeated the pig mucosa. A greater proportion of drug release was noted in films that contained sweeteners. Eight weeks of stability testing revealed no discernible changes in the amount of drug, the PH of the surface, or the amount of drug released in vitro

**Shimoda H et al., (2009)** Created an oral thin film with microcrystalline cellulose (PEG 400) as the foundation material and the medication dexamethasone that dissolved quickly. After being stored for up to 24 weeks at 40°C and 75% humidity, this product has demonstrated outstanding consistency and stability. After submerging this film in distilled water, it fell apart in a matter of seconds.

**Patel R et al., (2009)** Created a thin film that dissolves in the tongue and is helpful in treating antiemetic medications. These films contain maltodextrin, which is employed as an excipient because of its outstanding film-forming properties and appealing flavour, and ondansetron with low viscosity HPMC E15.

**Nishimura M et al., (2009)** Created oral disintegrating films for the administration of opioid

analgesics or anti-cancer medications. Microcrystalline cellulose, polyethylene glycol, and hydroxypropyl methyl cellulose were used as the basis ingredients, and prochlorperazine was present. When kept at 400C and 75 percent humidity, the film demonstrated outstanding stability for at least eight weeks. The dissolution test showed that prochlorperazine dissolved mostly in two minutes after being inserted into the liquid, indicating a rapid disintegration property.

**Aditya D et al., (2008)** An evaluation was conducted on several polyhydric alcohols, film modifiers, and film forming agents to enhance the composition of films with rapid dissolution. An investigation was conducted to determine the potential of poloxamer 407 and hydroxypropyl-β-cyclodextrin (HPBCD) to enhance the solubility of TC. Fast-dissolving films were fabricated by combining HPMC, xanthan gum, and xylitol. The utilization of poloxamer 407 with HPBCD resulted in a notable enhancement in the solubility of TC. The study involved the evaluation of the in vitro dissolution profile and in vitro microbiological studies of rapid dissolving films containing TC-HPBCD complex and TC-Poloxamer 407. The films comprising TC-poloxamer 407 exhibited superior in vitro dissolution profile and in vitro microbiological study in comparison to the films containing TC-HPBCD complex. An evaluation was conducted on human volunteers to determine the influence of eugenol inclusion on the in vivo efficacy of films containing TC-Poloxamer 407. The use of eugenol in TC-Poloxamer 407 films resulted in improved acceptance in terms of flavor masking and tongue refreshing, while still maintaining the in vivo disintegrating duration.

**Francesco Cilurzo et al. (2008)** The investigation focused on the utilization of maltodextrins with a low dextrose equivalent as film-forming agents for the development of oral fast-dissolving films. The selection of the suitable plasticizer and its concentration was based on the flexibility, tensile strength, and stickiness of the MDX films. The interactions between the MDX and plasticizer were investigated using ATR-FTIR spectroscopy. Flexible films were fabricated using a concentrate of 16-20% w/w glycerin. The principal production procedures, namely casting and solvent evaporation (Series C) or hot-melt extrusion (Series E), were modified by incorporating sorbitan monooleate or microcrystalline cellulose into the fundamental formulation. The presence of microcrystalline cellulose had a significant effect on both the disintegration time and ductility of the film, as observed in both in vitro and in vivo experiments (Series C 10s; Series E 1min). The film loading capacity was assessed using piroxicam (PRX), a medicine that is insoluble in water. When a drug was loaded as a powder, the film ductility decreased. However,

the formulation still maintained sufficient flexibility and resistance to elongation for manufacturing and packing purposes. The films possess a substantial loading capacity of up to 25 mg, owing to their surface area of 6 cm2. Irrespective of the PRX/MDX ratio, there was a significant rise in the dissolution rate of PRX in Series C films.

**Mashru R.C. et al (2005)** Created a salbutamol sulphate film that dissolves quickly, which is helpful during an asthma attack. The film is administered sublingually and was made by a solvent evaporation process. Mannitol serves as a filler, glycerol serves as a plasticizer, and polyvinyl alcohol serves as a polymer in the movie. It was discovered that the best response values for fast-release film could be achieved at moderate concentrations of glycerol and polyvinyl alcohol combined with a high concentration of mannitol.

**Masao et al., (2004)** glucomannan (1:2 mixture of glucose and mannose with acetyl and phosphate groups producing pendant ester linkage), natural polysaccharide (Pullulan, carrageenan, cellulose, xanthan gum), and polyhydric alcohol (glycerin) to create patentable edible films that are heat- and water-resistant. In the presence or absence of alkali, glucomannan was combined with Pullulan, another naturally occurring polysaccharide with numerous hydroxyl groups and polyhydric alcohols. Water-resistant films were created as glucomannan and natural polysaccharides interacted.

The Promethazine hydrochloride Fast-dissolving Oral Film was developed and evaluated by **AnjumPathan et al. (2016).** The films were fabricated by the process of solvent casting, employing PEG400 as a softener and HPMC E15 as the synthetic polymer foundation for the film. As surfactants, various quantities of microcrystalline cellulose (MCC) and Sodium Lauryl Sulphate (SLS) are utilized. The beverage has strawberry flavors and sugar sweetening. The films were deemed satisfactory based on evaluations of their thickness, weight uniformity in-vitro drug release, folding resilience, drug content, and disintegration time. The in vitro drug release of the improved formulation F2 was found to be 14.36% within a time frame of less than two minutes. Furthermore, the improved formulation F2 exhibited satisfactory stability, with a disintegration time of 09 seconds, an efficient in vitro drug release of 96.03±0.68% throughout a 16-minute timeframe, an appropriate pH environment, and a drug content of 97.41±0.54%.

**Safila Naveed et al., (2015)** examined the perindopril technique development. It is an inhibitor of the angiotensin-converting enzyme (ACE). Using water as a solvent, the absorbance at the

wavelength of perindopril's maximum absorption is measured in this approach. With a correlation coefficient of 0.9991, the calibration curve for perindopril was linear within the concentration range of 12.5-200 µg/ml. The method's precision and accuracy were established and statistically evaluated. Good recovery with % RSD less than 2 and good reproducibility were demonstrated by the approach. The technique was discovered to be quick, precise, accurate, and specific. Regular analysis of perindopril can be successfully performed using this approach. The suggested method's correctness was verified by accuracy studies that produced results that were approved. The suggested method's precision was verified by intraday and interday precision tests. In compliance with ICH recommendations, the perindopril determination method was validated.

**Smita V. Pawar, et al., (2015)** By using the solvent-casting process, prepared risperidone fast-dissolving films with propylene glycol serving as the plasticizer and HPMC E5 as the polymer were created. All of the created films had a similar thickness, weight, and medication content and appeared smooth and elegant with no obvious flaws. Formulation A2 is considered the optimum formulation due to its high percentage of elongation (120%), excellent folding endurance (185), rapid disintegration rate (13 seconds), and maximum in vitro drug release (93.57%) within less than 10 minutes. The stability tests conducted on the optimized formulation did not reveal any noticeable changes. The findings of the study suggest that the development of fast-dissolving oral films containing risperidone could be a promising therapeutic approach for schizophrenia. These films have the potential to offer improved bioavailability, a faster onset of action, and enhanced patient adherence.

The Quick Dissolving method was invented by **VasaviGeedi et al. (2015).**Oral films of zolmitriptan were fabricated by the solvent casting method, employing natural polymers like xanthan gum, guar gum, sodium alginate, and aloe vera powder as the film-forming polymer, and PEG-400 as the plasticizer. Vanillin was employed as a flavor-masking component in the recipe. The selected dependent variables included drug content, disintegration time, folding durability, thickness, and percentage of drug released. The optimized formulation, F6, was found to have superior performance compared to the other seven batches. Formulation F6 exhibits the lowest disintegration time of 8.33±0.57 seconds and the highest drug release of 99.1±0.05% within an 8-minute timeframe, owing to the super-disintegrant properties of sodium starch glycolate. Consequently, the films that contained 2%w/v sodium alginate and PEG-400 exhibited favorable

film-forming characteristics and a rapid drug release profile. FT-IR tests indicated the absence of any physico-chemical interaction between the medicine and polymer. The stability of the optimized formulation was determined using stability testing, which demonstrated a 99.1% drug release rate at the conclusion of the third month.

# AIM & OBJECTIVE

# CHAPTER-III

# AIM & OBJECTIVE

## Aim and Objective:

One of the main risk factors for peripheral vascular disease, heart failure, coronary artery disease, stroke, vision loss, and chronic kidney disease is high blood pressure. The goal of the current study was to create an oral disintegrating film that included the medication MXD, which is used to treat moderate to severe hypertension. MXD has a half-life of 2.4–2.6 hours and an oral bioavailability of only 80%. Therefore, it is a perfect choice to be formulated into films that dissolve when swallowed in order to effectively treat hypertension.

# Plan of work:

# CHAPTER-IV

## Plan of work:

- To carry out investigations for Preformulation.
- Using FTIR research to determine the drug-excipient compatibility
- Creation of Orally dissolving films using the Cast-solvent technique.
- Polyhydric alcohols and film forming agents are used to optimize films.
- The films are assessed based on multiple factors, including weight, thickness, Tensile strength, folding endurance, pH, hydration, and moisture loss.
- To evaluate how different formulation factors affect the permeability profile of the membrane.
- To determine the kinetics and mechanics of drug release from various formulations.

## ❖ Plan of work representation:

- ❖ Preformulation studies
  - ✓ Calibration of standard curve
  - ✓ FT-IR compatibility
  - ✓ Solubility studies
- ❖ Formulation of films
- ❖ Preparation of mouth dissolving films using a different polymers
- ❖ Characterization of films
  - ✓ SEM
  - ✓ Thickness
  - ✓ Folding endurance
  - ✓ Weight variation
  - ✓ Surface pH
  - ✓ Drug content uniformity
  - ✓ Disintegration time
- ❖ In vitro drug release kinetics

# DRUG PROFILE

## CHAPTER 1V

## DRUG PROFILE

**DRUG PROFILE: MOXONIDINE**

**CHEMICAL STRUCTURE:**

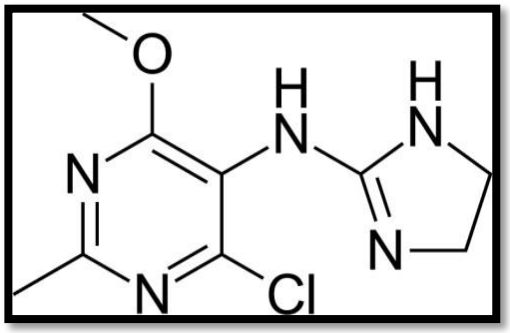


Figure 4: Structure of MXD

**IUPAC NAME** : 4-Chloro-N-(4,5-dihydro-1H-imidazol-2-yl)-6-methoxy-2-methylpyrimidin-5- amine

**Formula** : $C_9H_{12}C_lN_5O$

**Molar mass** : 241.7 g/mol

**Description** : white or almost white powder

**Solubility** : Very slightly soluble in water, sparingly soluble in methanol, slightly soluble in methylene chloride, very slightly soluble in acetonitrile.

**Pharmacokinetic data**

**Bioavailability** :88%

**Metabolism** : Hepatic(undergoes first pass metabolism)

**Biological half-life** : 2.2-2.6 hours

**Mechanism of action:** MXD selectively binds to the subtype I imidazoline receptor (I1). The medulla oblongata has this receptor subtype in both the ventromedial depressor and rostral ventro-lateral pressor regions. For this reason, MXD lowers blood pressure by decreasing

sympathetic nervous system activity. MXD has a significantly higher affinity for the imidazoline I1-receptor than the α2-receptor when it comes to central-acting antihypertensives than those that are older. Clonidine, on the other hand, binds to both receptors as well. Furthermore, MXD may enhance insulin resistance and glucose tolerance, increase sodium excretion, and guard against hypertensive target organ damage, including renal disease and heart hypertrophy.

**Side effects:** dry mouth; headache; exhaustion; nausea; dizziness; headaches; asthenia; vasodilatation; seldom, skin reactions; extremely uncommon angle closure glaucoma

**Uses:** used to treat essential hypertension that is mild to moderate.

# EXCIPIENT PROFILE

# CHAPTER VI

# EXCIPIENT PROFILE

## POLY ETHYLENEGLYCOL:

**Nonproprietary Names** **:** Macrogols

**Synonyms** **:** Carbowax; Carbowax Sentry; Lipoxol; Lutrol E; PEG; Pluriol E; poly oxyethylene glycol.

**Chemical Name** **:** α-Hydro-ω-hydroxypoly(oxy-1,2-ethanediyl)

**Empirical Formula** **:**$HOCH_2(CH_2OCH_2)_nCH_2OH$

**Description :**

Polyethylene glycol is defined as an addition polymer of ethylene oxide and water in the USPNF 23. At room temperature, polyethylene glycol grades 200–600 are liquids, while grades 1000 and higher are solids. PEG 200–600 liquid grades are transparent, colourless, or somewhat yellow-colored, viscous liquids. They taste harsh and slightly burnt, with a subtle yet distinct odour. PEG 600 can exist at room temperature as a solid. Solid grades (PEG>1000) come in a variety of consistency ranges, from pastes to waxy flakes, and are white or off-white in colour. They smell slightly delicious. PEG 6000 and higher grades are offered as free-flowing, milled powders.

Table 12: Molecular weight of typical polyethylene glycol

| Grade | Average Molecular weight |
|---|---|
| PEG 200 | 190-210 |
| PEG 300 | 285-315 |
| PEG 400 | 380-420 |
| PEG 540 | 500-600 |
| PEG 600 | 570-613 |
| PEG 900 | 855-900 |
| PEG 1000 | 950-1050 |
| PEG 1450 | 1300-1600 |

| | |
|---|---|
| PEG 1540 | 1300-1600 |
| PEG 2000 | 1800-2200 |
| PEG 3000 | 2700-3300 |
| PEG 3350 | 3000-3700 |
| PEG 4000 | 3000-4800 |
| PEG 4600 | 4400-4800 |

**Functional Category :** basis for ointment; plasticizer; solvent; basis for suppository; lubricant for tablets and capsules.

**Structure :**

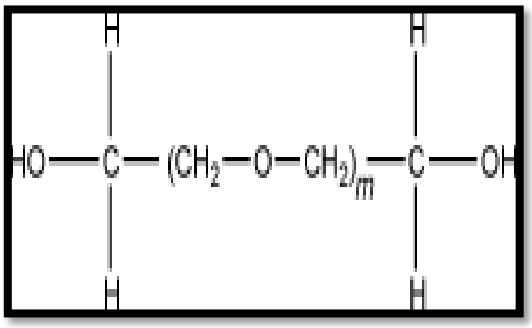


Figure 5: Structure of Macrogols

**Solubility :**

Every grade of polyethylene glycol has a water solubility and can mix with other grades of polyethylene glycol in any ratio (after melting, if necessary). Gels can be formed by aqueous solutions with greater molecular weight classes. Glycerin, benzene, acetone, and alcohols are all soluble in liquid polyethylene glycols. Solid polyethylene glycols are insoluble in mineral oil, fats, fixed oils, and acetone, but soluble in methanol, ethanol (95%), and dichloromethane. They are also somewhat soluble in ether and aliphatic hydrocarbons.

**Applications in Pharmaceutical Formulation or Technology:**

In many different pharmaceutical formulations, such as parenteral, topical, ophthalmic, oral, and rectal preparations, polyethylene glycols, or PEGs, are commonly utilized. It has been tested in controlled-release systems using biodegradable polymeric matrix.

Stable and hydrophilic, polyethylene glycols are basically non-irritating to the skin.

Although the polyethylene glycols are water soluble and easily washed off the skin, they do not easily enter the skin, which makes them ideal as bases for ointments.

Topical ointments typically use solid grades of polyethylene glycol, while liquid grades are added to modify the base's viscosity.

One can utilize aqueous polyethylene glycol solutions as suspending agents or to modify the consistency and viscosity of other suspending vehicles. Polyethylene glycols can function as emulsion stabilizers when combined with other emulsifiers. The water-miscible solvents employed for the soft gelatin capsule contents are liquid polyethylene glycols. However, through preferential absorption of moisture from the gelatin in the shell, they may cause the capsule's shell to harden.

**HYDROXY PROPYL METHYL CELLULOSE:**

**Non proprietary names :-** Hypermellose, Hydroxy propyl methyl cellulose

**Synonyms:** Methylhydroxylpropylcellulose, Propylglycoletherofmethylcellulose, Methocel, Benecel MHPC, Pharmacoat.

**Chemical name :-** Cellulose, 2-hydroxy propyl methyl ether.

**Description:** HPMC is a non-ionic polymer that is white or creamy white, tasteless, odorless, and fibrous in powder form.

**Functional category:** HPMC is a non-ionic polymer that is white or creamy white, tasteless, odorless, and fibrous in powder form.

**Structure :**

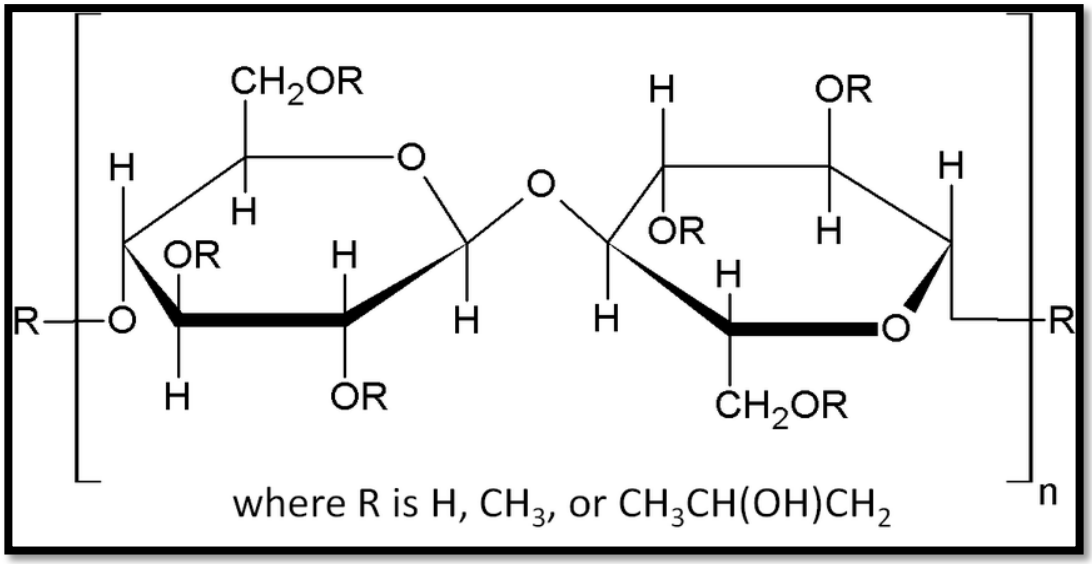


Figure 6: Structure of Hydroxy propyl methyl cellulose

**Solubility:** soluble in cold water, where it condenses as a thick colloidal solution. practically insoluble in ether, 95% ethanol, and chloroform. However, soluble in alcohol and water mixtures as well as ethanol and di-chloro methanol mixtures.

**Viscosity:** HPMC E3-3 (m pas) HPMC E6-6(m pas)

HPMC E15-15(m pas)

**Applications:**

A common ingredient in medicinal formulations is HPMC. HPMC is utilized in oral goods as a wetting agent for hard contact lenses, a tablet binder, a film coating agent, and a thickening ingredient in topical formulations, especially ophthalmic treatments. used to slow down the drug's release profile.

**XANTHAN GUM**

**Non-proprietary Names:** Xanthum Gum, Xanthani gumani, Xanthun gum

**Synonyms:** Corn sugar gum; E415; Keltrol; polysaccharide B-1459; Rhodigel; Vanzan NF; Xantural.

**Empirical Formula:** $(C_{35}H_{49}O_{29})_n$

**Molecular Weight:** Approximately 2 x$10^6$

Xanthan gum is classified as a high molecular weight polysaccharide gum in the USPNF 23. It is made as the sodium, potassium, or calcium salt and has D-glucose, Dmannose, and D-glucuronic acid as the main hexose units.

**Structural Formula:**

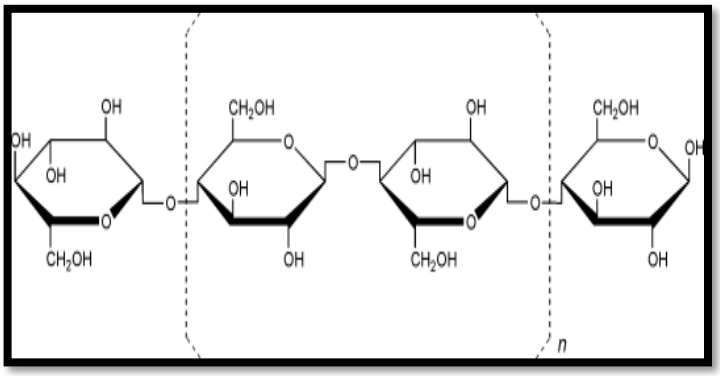


Figure 7: Structure of Xanthan Gum

Five sugar residues—two glucose, two mannose, and one glucuronic acid—are present in each repeat unit of xanthan gum. The polymer backbone is the same as cellulose since it is made up of four $\beta$ -D-glucose units connected at the 1 and 4 locations. Xanthan is distinguished from cellulose by alternating anhydroglucose units with side chains of trisaccharides. A glucuronic acid residue sits between two mannose units in each side chain. A pyruvate moiety is present in the majority of terminal mannose units; the mannose closest to the main chain only has one group, at C-6. The resultant stiff polymer chain interacts with other xanthan gum molecules to build intricate, loosely connected networks. It can exist in solution as a single, double, or triple helix.

**Description:** Xanthan gum is found as a fine, odourless, cream- or white-colored powder that flows freely.

**Functional Category**

- Stabilizing agent
- Suspending agent
- Viscosity-increasing agent

**Applications:**

Xanthan gum is a common suspending and stabilizing ingredient in foods, cosmetics, and oral and topical medicinal formulations. It is also employed as an emulsifying and thickening agent. It has good stability and viscosity qualities across a broad pH and temperature range, is nontoxic, and is compatible with the majority of other medicinal components. Pseudo plastic behavior is shown by xanthan gum gels, where shear thinning is directly proportional to shear rate. The instant the shear tension is released, the viscosity returns to normal. Synergistic rheological effects happen when xanthan gum is combined with some inorganic suspending agents, like magnesium aluminium silicate, or organic gums. The best results are often obtained from xanthan gum and magnesium aluminium silicate combinations in ratios between 1:2 and 1:9. Similarly, ratios of xanthan gum to guar gum between 3:7 and 1:9 yield the most synergistic effects. Xanthan gum has been utilized to make sustained-release matrix tablets, despite its primary application as a suspending agent. It has been claimed that xanthan gum is used to manufacture controlled-release diltiazem hydrochloride tablets that maintain drug release in a predictable way, with no effect from pH or agitation rate on the drug release profiles of these tablets. An ophthalmic liquid dosage form that contains xanthan gum interacts with mucin to help prolong the dosage form's retention in the precorneal region. According to recent research, xanthan gum can also be employed as an excipient to improve the outcomes of freeze- and spray-drying procedures. Xanthan gum can be utilized as a binder in colon-specific drug delivery systems and to boost the bioadhesive strength of formulations intended for vaginal use. In the food business, xanthan gum is also utilized as a hydrocolloid; in cosmetics, it is employed as a thickening ingredient in shampoo.

**Typical Properties:**

- Acidity/alkalinity: pH = 6-8 (1% w/v aqueous solution)
- Melting point: Chars at270°C
- Solubility: Practically insoluble in ethanol and ether; soluble in cold or warmwater.
- Specific gravity: 1.600 at250°C
- Viscosity (dynamic): 1200-1600mPas (1200-1600 cP) for 1% w/v aqueous solution at 250°C.

**Method of Manufacture**: A polysaccharide known as xanthan gum is created when Xanthomonascampestris ferments a carbohydrate in a pure culture. After recovery with propan-2-ol, the polysaccharide is dried and ground.

**Stability and storage condition:** A substance that is stable is xanthan gum. Aqueous solutions exhibit stability throughout a broad pH range of 3 to 12 and a temperature range of 10 to 600°C. Temperatures higher than the surrounding air can have a negative impact on xanthan gum solutions with less than 1% w/v concentration.

# MATERILAS & METHODOLOGY

## CHAPTER VII

# MATERIALS & METHODOLOGY

TABLE NO: 11 LIST OF CHEMICLAS USED:

| Sl. No. | INGREDIENTS | SOURCES | CATEGORY |
|---|---|---|---|
| 01. | Moxonidine | Gift sample from UniChem laboratories Ltd., Mumbai | Active ingredient |
| 02. | HPMC | Colorcon Asia Pvt. Ltd., | Polymers/film Forming agents |
| 03. | Xanthan gum | Shilex Chemicals | Polymers/film Forming agents |
| 04. | Methanol | Merck | Solvent |
| 05. | Dichloro methane | S.D. Fine Chem Ltd., | Solvent |
| 06. | Poly ethylene glycol | S.D. Fine Chem Ltd., | Plasticizer |
| 07. | Aspartame | HI media Lab Pvt Ltd., | Sweetener |
| 08. | Orange flavor | Pentagon trading company | Flavoring agent |
| 09. | Disodium hydrogen phosphate | Merck Ltd, Mumbai. | For Buffer Preparation |
| 10. | Sodium hydroxide | S.D. Fine- Chemical Ltd, Mumbai | For Buffer Preparation |

Table 12: List of Equipments

| S. No. | Equipments | Source |
|---|---|---|
| 1. | Electronic weighing balance | AW 120, Shimadzu Corporation, Japan |
| 2. | Dissolution apparatus | Electro lab TDT- 08L, Mumbai, India. |
| 3. | UV-VIS spectrophotometer | Elico, Ahmedabad, India. |
| 4. | Digital Vernier calipers | Mitutoyo corporation, Japan. |

## PREFORMULATION STUDIES

The phase of development known as preformulation is when a drug substance's physicochemical and biologic characteristics are determined. It is a crucial step in the creation of new drugs. Critical decisions in subsequent stages of development are made using the drug development knowledge obtained during this phase. To construct formulations rationally, a wide range of information must be generated. Studying the qualities of the excipients and their compatibility with the medicine is a crucial step in the preformulation phase of product development.

### Solubility

Parts per million of solvent in which one gramme of substance is soluble is the unit of solubility expression. At 20°C, the powder's solubility in several solvents, such as water and ethanol, was assessed.

### Melting point

The capillary tube method was utilised to determine the melting point.

### Construction of calibration curve of MXDby UV spectrophotometry

### Preparation of pH 6.8 phosphate buffer (IP 2007 vol 1)

In enough water to provide 1000 millilitres, dissolve 28.8 grammes of disodium hydrogen phosphate and 11.45 grammes of potassium dihydrogen phosphate.

### Calibration graphs preparation in Phosphate buffer pH 6.8

100 milligrammes of the medication alone After precisely weighing the morphonidine, it was transferred to a 100 ml volumetric flask. Phosphate buffer 6.8 was used to dissolve the drug, and 100 millilitres was the final volume. The drug's concentration was 1000 µg/ml.

### Standard solution

In a 100 ml volumetric flask, 100 mg of MXD was dissolved, and water was added to get the solution to the proper level.

**Calibration ofMXD**

One hundred milligrammes of MXD precisely weighed, is added to a 100 millilitre volumetric flask. Using 6.8 pH phosphate buffer, which is the stock solution of 1 mg/ml, the volume was increased to 100 ml. Solutions with concentrations of 5, 10, 15, 20, 25, and 30 µg/ml were made by further diluting the stock solution appropriately (0.5, 1, 1.5, 2, 2.5, and 3 ml are taken from stock solution and dilute with 100 ml buffer) using 6.8 pH phosphate buffer. A standard graph was plotted and the absorbance of these solutions was measured using a UV-VIS spectrophotometer set to λmax 249 nm, with a blank of 6.8 pH phosphate buffer.

**Fourier Transform Infrared (FT-IR) Spectroscopy**

The drug's FT-IR spectra was captured with a JASCO 4100. The mid-IR 4000-400 cm-1 spectral range was covered by the diffuse reflectance approach. The process entails utilizing a motor to disperse the sample in kbr (100 mg), then a compression gauge to triturate the materials into a fine powder bed inside the holder. For five minutes, the pressure was about five tons. The pellet was positioned along the light path, the spectrum was captured twice, and the distinctive peaks of the functional groups were deciphered.

**UV Spectroscopy**

Ten milliliters of water were added to a precisely weighed dose of 10 milligrams of medication. Phosphate buffer solution was made using 20 µg/ml of MXD solution from the principal stock solution. The range of 200–400 nm was scanned with this solution.

**Formulation Design:**

Using the solvent casting method, MXD ODFs were made with several grades of Hydroxy Propyl Methyl Cellulose, such as HPMC E3, HPMC E6, and a natural polymer called Xanthum gum. We set a formula with the following components:

| | | |
|---|---|---|
| Drug | : | Moxonidine |
| Solvents | : | Methanol and Dichloro methane |
| Polymers/Film forming agent | : | HPMC E3, HPMC E6 & Xanthan gum |
| Plasticizer | : | PEG400 |
| Sweetener | : | Aspartame |

Flavoring agent : Orange flavor

The formulae of different formulations are as follows:

Table 13: Formulation of Moxonidine ODF

| **Ingredients (mg)** | **MC1** | **MC 2** | **MC 3** | **MC 4** | **MC 5** | **MC 6** | **MC 7** | **MC 8** | **MC 9** | **MC 10** | **MC 11** | **MC 12** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Moxonidine chloride | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| HPMC E3 | 50 | 100 | 150 | -- | -- | -- | -- | -- | -- | -- | -- | 50 |
| HPMC E6 | -- | -- | -- | 50 | 100 | 150 | 200 | -- | -- | -- | -- | 150 |
| Xanthan Gum | -- | -- | -- | -- | -- | -- | -- | 50 | 100 | 150 | 200 | 200 |
| PEG 400 | 40 | 40 | 40 | 40 | 40 | 40 | 40 | 40 | 40 | 40 | 40 | 40 |
| Aspartame | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 |
| Orange Flavor | Q. S | Q. S | Q. S | Q. S | Q. S | Q. S | Q. S | Q. S | Q. S | Q. S | Q. S | Q. S |

* Processing solvents Dichloro methane, Methanol in the preparation of films, not present in the final formulation

**Preparation of ODF**

**Method Employed:** Solvent Casting Method

**Procedure:** With the use of xanthan gum, HPMC E3, and HPMC E6, the ODF of MXD was created. Dichloromethane and methanol were combined in a 1:1 ratio to create the polymeric solution, which was then left for five to six hours to allow the polymer to swell. After MXD was dissolved in a specific amount of solvent, the drug solution was added to the polymeric solution mentioned earlier. The inclusion of plasticizers, such as PEG 400, came next. Flavor and sweetener were also added. The medication content can be made uniform by mixing for 15 to 20 minutes in a cyclo mixer. The mixture was placed into a ready mold and let to air dry for forty-five minutes. After the film was meticulously taken out of the mold, it was inspected for flaws and trimmed to the proper size so that each strip would have the appropriate dosage (2 × 2cm2). Film samples that had cuts, air bubbles, or other flaws were not included in the investigation.

Distilled water was used to dissolve the plasticizers and water-soluble polymers. To release all trapped air bubbles, the solution is agitated for two hours in a magnetic stirrer and then left aside. In the meantime, the medication and excipients were dissolved and vigorously agitated for 30 minutes. Both solutions were then combined after the stirring was finished. In order to create a film, the solution is finally cast onto an appropriate Petri plate. For one hour, the plates were stored at 60°C in a hot air oven. After drying, the film was carefully cut to the appropriate size and removed from the glass plate.

**Dose calculations**

Length of glass plate =10 cm.

Width of glass plate =10 cm.

Area of the plate =100 cm2.

No. of 4 cm2 films present whole plate =100/4 =25 films.

Each films contains 0.2 mg of drug.

25 films contain 5 mg drug (25×0.2).

Labelled claim= 0.2mg

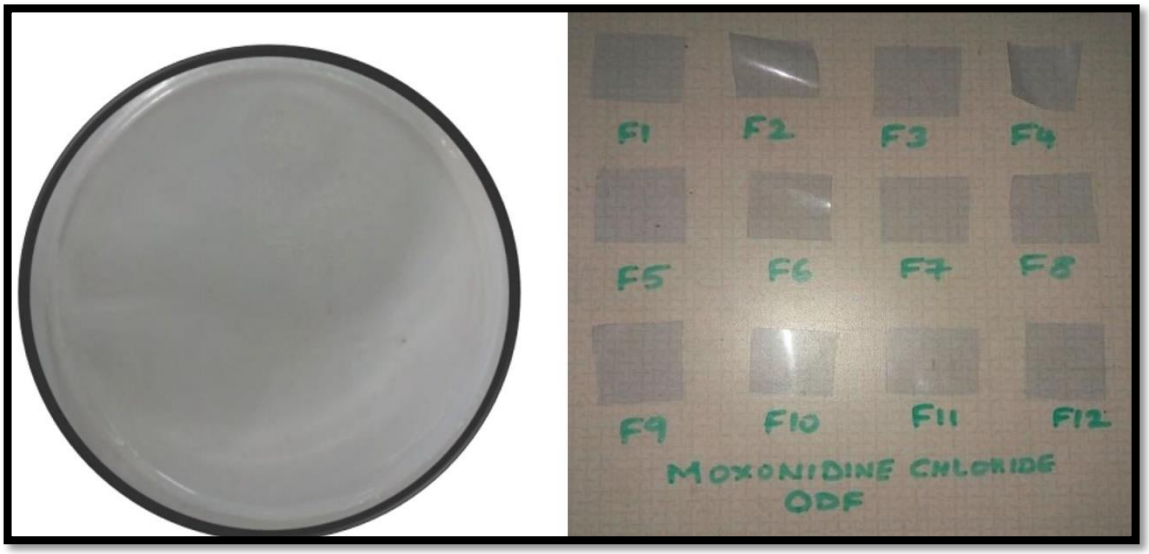


Figure 8 : Orally Disintegrating films of MXD

**Drug - Polymer Compatibility Studies**

It is crucial that a drug material be compatible both chemically and physically before it is formulated into a dosage form. When a drug is combined with pharmaceutical excipients to create a dosage form, compatibility studies provide the framework for the combination and the information required to characterize the nature of the drug substance. Compatibility is one of the criteria for choosing appropriate excipients or carriers for pharmaceutical formulation. Consequently, an investigation was conducted in the current work utilizing an infrared spectrophotometer to determine whether MXD and excipients could potentially interact chemically. A mixture of 100 mg of potassium bromide (dried at 40-50˚C) and 3 milligrams of the medication was weighed.

To create a transparent pellet, the mixture was taken and compressed in a hydraulic press at a pressure of 10 tons. The particle was scanned in an infrared spectrophotometer between 4000-400 cm-1.

To determine whether the medication and polymers were compatible, an FTIR research was conducted. MXD's infrared spectra was identified using the Fourier transform. KBr dispersion technique infrared spectrophotometer. With dried potassium bromide, the baseline correlation was carried out. Next, using an FTIR spectrophotometer, the spectra of a dried mixture of the drug and potassium bromide was run, and then the drug with other polymers. The position and relative strength of the absorption maximums in the spectrum produced with the substance under examination match those in the reference spectrum. The careful selection of excipients, which are included to aid in administration, support the drug's constant release and bioavailability, and shield it from degradation, is essential to the formulation of a stable and effective solid dosage form. Compatibility studies are crucial if the excipients are novel and haven't been used in a formulation that contains the active ingredient. MXD’ s compatibility with Xanthan Gum, PEG 400, and HPMC was determined by Fourier Transform Infrared Absorption Spectral Analysis (FTIR).

**FT-IR peak matching**

The method of matching FT-IR spectra of the FT-IR data was used to find the chemical interaction between the medication and particular polymers. Changes in the peak (characteristic wave numbers) can be used to identify the chemical interaction between the medication and polymer. An appropriate amount of potassium bromide was individually mixed with a physical

mixture of the drug, polymer, and drug & polymer. This sample weighed around 100 mg, and it was compressed with 15 tons of pressure in a hydraulic press to create a translucent pellet. The scanning range was 4000-400/cm. The FT-IR spectra of the drug formulation, polymer, and pure drug were compared to look for peak presence or disappearance.

**Evaluation of oral dissolving films :**

For ODF formulations, various quality control tests were carried out.

- Measurement of thickness,
- Assay
- Tests for in vitro dissolution
- Weight fluctuation
- Folding endurance
- Surface pH
- Tensile strength
- Percentage elongation

**Thickness measurement:**

The digital vernier caliper was used to measure the thickness and size (diameter). The film's thickness was measured five times using a micrometer screw gauge, and an average of three readings was determined. Ensuring consistency in the thickness of the film is crucial as it directly affects the accuracy of the dose within the film. Bande, Swapnali V. et al., 2007. A micrometer screw gauge was employed to determine the thickness of the film. The thickness of the film is measured five times to ensure uniformity. The film is Less than 5% should be the thickness.

Figure 9: digital vernier caliper

**Method:** Five points on the film should be measured: the center, the four corners, and the mean thickness. Six films of each formulation should be used for this test. The value was stated in millimeters.

**Weight variation:**

A weight was determined by selecting ten films at random and averaging them. Weighing each film, we compared its weight to the deviation's average. Using an analytical balance, the average weight of the mouths dissolving oral films was calculated. It is preferable if the weight of films is almost consistent. Making sure a film has the right amount of API and excipients is helpful.

**Folding endurance:**

A film is sliced and rapidly folded in the same spot until it breaks to test folding endurance. The value of folding endurance is determined by the number of times the film could be folded in the same way without breaking. Film had a folding endurance of 100–150.

**Surface pH**

The film that was going to be tested was put in a Petri dish, wet with 0.5 milliliters of distilled water, and left for thirty seconds. After allowing one minute for equilibration and contacting the formulation's surface with the pH meter's electrode, the pH was recorded. For every formulation, an average of three determinations was made. (Kumar GV and others, 2005)

**Assay of the Films:**

The drug content of the prepared Oro dissolving films was tested.

**Method:**

One film, chosen at random from the five, was weighed, then added to 100 milliliters of 6.8 pH in a volumetric flask. For thirty minutes, a volumetric flask was submerged in a sonicator. The final solution's absorbances were measured at 249 nm wave length using a UV Visible spectrophotometer against a blank of 0.01N HCl. Using a conventional graph, concentrations were computed, and the formulation's total amount was determined.

$$\text{Drug content} = \frac{\text{sample absorbance}}{\text{standard absorbance}} \times \frac{\text{standard dilution}}{\text{sample dilution}} \times \frac{\%\text{purity of drug}}{100} \times \text{Avg.wt}$$

$$\%\ \text{Drug content} = \frac{\text{Drug content} \times 100}{\text{Label claim}}$$

**Percentage elongation:**

It was calculated by

$$\text{Percentage elongation} = \frac{\text{Increase in length of strip} \times 100}{\text{Initial length of strip}}$$

**Tensile strength:**

The greatest stress applied to the point at which the strip specimen breaks is known as its tensile strength. The formula is used to calculate it.

$$\text{Tensile strength} = \frac{\text{Load at failure} \times 100}{\text{Strip thickness} \times \text{strip width}}$$

## Scanning electron microscope

With the use of scanning electron microscopy, the surface morphology of the optimized formulation was investigated. A double-sided tap-equipped scanning electron microscope sample holder was coated with 150˚A gold for two minutes using a JSM 6390 sputter coater (made by JEOL) in a 3×10-1atm organ gas vacuum. A scanning electron microscope was then used to investigate the materials (Gupta Dilip Kumar et al., 2014)..

**In vitro Dissolution test:**

**Method:**

The developed ODF formulations were subjected to an in-vitro dissolving investigation utilizing an Electro lab dissolution rate test instrument in a USP type I (basket). ODFs with the appropriate formulation were taken and put inside the dissolution apparatus's vessels. Samples

were taken out of the vessels at various intervals, replaced with an identical volume of the blank solution, and subjected to UV-V is spectrophotometer analysis. Drug concentration was determined using the standard graph and reported as a percentage of the drug that was released or dissolved. Six duplicates of the release studies were carried out, and mean values were recorded.

**Conditions for dissolution of MXD ODFs:**

| | | |
|---|---|---|
| Apparatus | : | Dissolution Test apparatus (basket, modified USP I) |
| Medium | : | 500 ml, 0.01 NHCl |
| RPM | : | 50 |
| Temperature | : | 37 ± 0.5 $^{0}$C Sampling |
| Volume | : | 5 ml |
| Sampling intervals | : | 2, 4, 6, 8, 10, 12, 14, 16, 18, and 20 minutes |

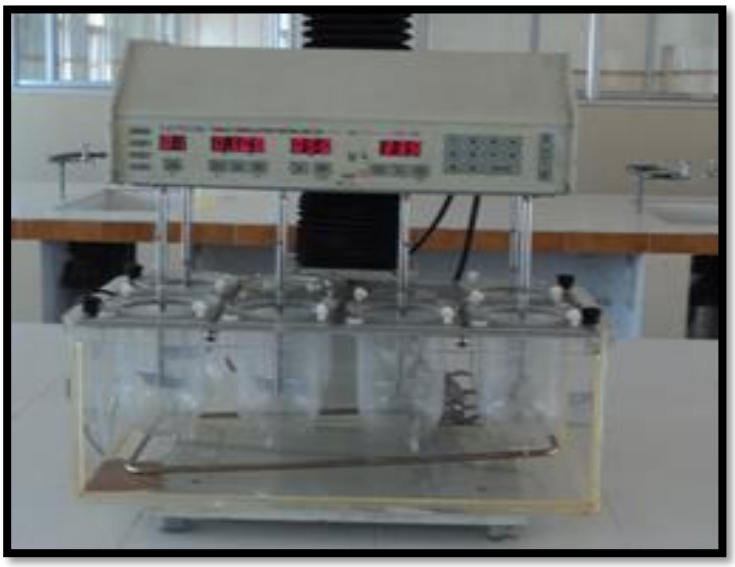

Figure 10: Dissolution Test Apparatus

**Release Kinetics**

The findings from the invitro diffusion investigation were used to investigate the order and mechanism of drug release kinetics of MXD films. The kinetic models that were plotted included the zero order, first order, and Higuchi equations; the release was calculated using the Korsmeyer-Peppas equations. The acquired data were plotted as follows in order to examine the mechanism of the dosage form's drug release rate kinetics Drug release cumulative percentage against time (also known as zero order or in-vitro drug release plots) Record the total proportion

of medicine left versus time using first-order charts.

**Zero-Order Kinetics:**

Plotting the total amount of medication released against time.

$$C = K_0 t$$

where t is the time in hours and K0 is the zero-order rate constant given in concentration/time units. A concentration vs. time graph would show a straight line with an axis origin intercept and a slope of K0. The formulations' concentration-independent drug release is described by this kinetics.

**First order kinetics:**

Log cumulative proportion of medicine remaining vs. time is the first order of order. The formulations' concentration dependent drug release is described by this kinetics.

$$\text{Log } C = \text{Log } Co\text{-}kt / 2.303$$

where t is the time, k is the first order constant, and C0 is the drug's starting concentration.

**Higuchi's Model:**

The Higuchi model represents the drug release cumulative percentage as a function of time squared.

$$Q = Kt_{1/2}$$

where t is the time in hours and K is the constant representing the system's design variables. According to this model, the drug's release from a swellable matrix is described as a time-dependent process with a square root based on Fickian diffusion.

**Korsemeyer-Peppas Equations:**

The method of drug release involved plotting the first 60% of drug release as a log cumulative percentage of drug released vs. log time in Korsemeyer et al's equation, then calculating the exponent n using the slope of the straight line.

$$Mt / M_\infty = Kt^n$$

where t is the release time, K is a kinetic constant, and $M_t/M_\infty$ is the fractional solute release. the drug/polymer system characteristic, and n is an exponent that describes the tracers' release mechanism. When it comes to cylindrical matrix tablets, Fickian diffusion is the drug release mechanism if the exponent is 0.45; non-Fickian or anomalous diffusion occurs when $0.45 < n < 0.89$. According to Balaiah et al. (2012), an exponent value of 0.89 indicates a normal zero-order release or Case-II Transport.

# RESULTS AND DISCUSSION

# CHAPTER-V

# RESULTS & DISCUSSION

## Identification of MXD

### Solubility

Parts per million of solvent in which one gram of substance is soluble is the unit of solubility expression. At 20°C, the powder's solubility in several solvents, such as water and ethanol, was assessed. MXD has the following solubility levels: very faintly soluble in acetonitrile, sparingly soluble in methanol, somewhat soluble in methylene chloride, and somewhat soluble in water. Because the medication must dissolve in both the solvents and the dissolving medium utilized, solubility analysis is crucial.

### Melting point

We used the capillary tube method to measure the melting point. The table displays the determined melting point of MXD, which is 219ºC. A substance's melting point will change if there are any impurities present.

Table 14: Identification of MXD

| Sl. No. | Tests | Requirements | Results |
|---|---|---|---|
| 1 | Description | White, fine, crystalline powder | Complies |
| 2 | Solubility | Slightly soluble in water, sparingly soluble in methanol, slightly soluble in methylene chloride, very slightly soluble in acetonitrile. | Complies |
| 3 | Melting point | $219^0$C | 217-$225^0$C |
| 4 | Loss on drying | Not more than 0.5% t $105^0$C for 4 hrs | 0.2% |
| 5 | Sulphated ash | Not more than 0.1% | 0.06% |
| 6 | Assay | MXD contains not less than 95% and not more than 102.0% of calculated with reference to the dried substance | 99.17% |

**Calibration of MXD**

To make the stock solution, mix 50 mg of MXD with 100 ml of water. 10 milliliters of this stock solution were taken out and diluted with water to make 100 milliliters. A calibration curve was created using various concentrations (2–10 µg/ml) and the proper stock solution dilution. At 249 nm, the absorbance was measured. The table below shows the absorbance at different concentrations recorded at 249 nm. MXD was calibrated in a pH 6.8 pH phosphate buffer; the resulting curve is displayed in Figure 14. The outcomes were tallied in a table.

Table 15: Standard calibration curve of MXD in 6.8 pH phosphate buffer

| S. No | Concentration (µg/ml) | Absorbance in 6.8 pH buffer |
|---|---|---|
| 1 | 0 | 0 |
| 2 | 2 | 0.106 |
| 3 | 4 | 0.211 |
| 4 | 6 | 0.343 |
| 5 | 8 | 0.463 |
| 6 | 10 | 0.596 |

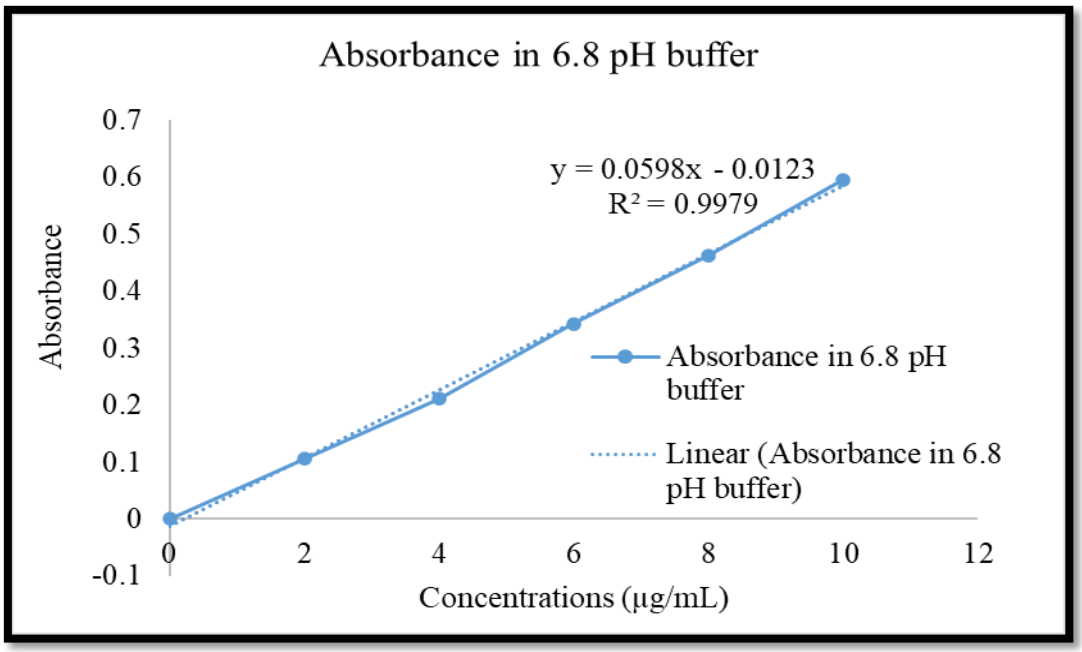


Figure 11: Standard Calibration Curve of MXD in 6.8 pH phosphate buffer

**Drug – excipient CompatibilityStudies**

FTIR spectroscopy was used to determine the drug excipient compatibility, and the graphs from the figure were displayed. To find out if there was any interaction between the excipients and MXD, the physical mixture was put through FTIR analysis. The lack of a drug-carrier chemical interaction is confirmed by the absence of any drug-characteristic peak appearance or disappearance. For the purpose of developing the formulation further, the excipients sodium alginate, xanthan gum, and pectin that were found to be compatible with MXD were chosen. MXD, HPMC, Xanthum gum, PEG 400, and HPMC E6 physical mixtures all had their Fourier transform infrared spectra recorded and examined for chemical interactions. All three samples, which were pure MXD, underwent FTIR analysis to determine the presence of the pure API in the mixtures and to describe it.

The acquired FTIR spectra are shown in Figure superimposed. The observations are consistent with the scientific literature, and the spectrum clearly displays the main characteristic bands linked to pure MXD. Therefore, for the principal amine groups, the band observed in the high wavenumber range (3300–3125 cm−1), with a maximum at 3187 cm−1, reveals the N–H stretching vibration. This vibration type's value is marginally below its typical position; the shift is most likely the result of hydrogen bond formation. The peak observed at 1648 cm−1 is linked to the bending vibration of the N–H bond, whilst the signals at 1219 and 1181 cm−1 are associated with the stretching vibrations of the several C–N bonds present in the molecule's architectural structure. The peaks observed at 1561 and 1524 cm−1 correspond to the stretching vibrations of the double C=N bonds found in the pyrimidine and 2-imidazoline heterocycles.

Given that the ether bond in the MXD structure produces a C–O–C linkage, whose stretching vibration defines a significant signal, it is easy to tie this bond's existence to the absorption band seen at 1092 cm−1. There is a noticeable peak at 857 cm−1, which most likely signifies the existence of the C–Cl bond. The remaining hydrocarbon skeleton exhibits a range of characteristic vibrations, including the bending vibrations of the –CH3 groups (1469 and 1369 cm−1), the scissoring (1426 cm−1), wagging (1313 cm−1), and rocking (705) vibrations of the –CH2 groups, and the symmetric and asymmetric stretching vibrations of the C–H bonds from the methyl and methylene moieties (2997, 2953, and 2856 cm−1).

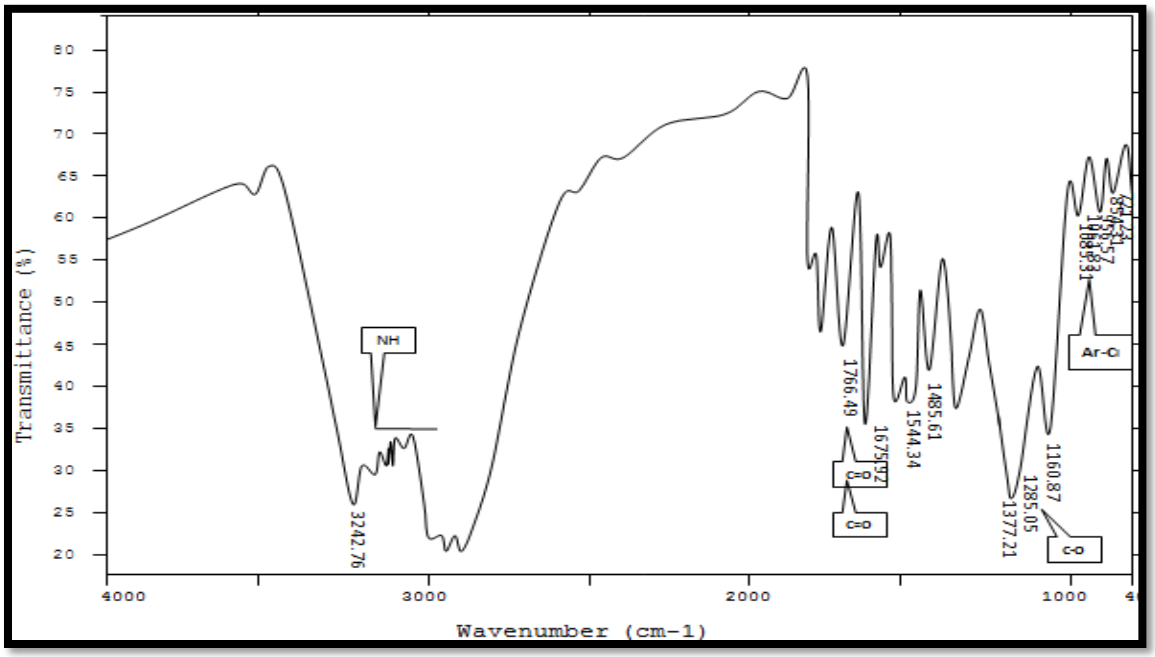


Figure 12: FTIR Spectrum of pure drug MXD

To determine whether the medication and polymers were compatible, an FTIR research was conducted. Using the KBr dispersion method, the Fourier transform infrared spectrophotometer was used to determine the MXD's infrared spectrum. With dried potassium bromide, the baseline correlation was carried out. Next, using an FTIR spectrophotometer, the spectra of a dried mixture of the drug and potassium bromide was run, and then the drug with other polymers. The position and relative strength of the absorption maximums in the spectrum produced with the substance under examination match those in the reference spectrum. Before the tablets were prepared, compatibility experiments between polymers and the pure medication MXD were conducted. The acquired I.R. spectra of the medication m Oxonidine chloride in its pure form and with the addition of excipients are shown. The presence of all five of the MXD characteristic peaks in the spectra at the appropriate wavelengths—3242.76 cm-1(-NH), 1766.49 cm-1, 1675.92 cm-1 (C=0), 1285.05 cm-1 (C-O), and 1061.83 cm-1 (Ar-Cl)—indicates that the drug and excipients are compatible. It demonstrates that the drug's chemical integrity has not changed significantly.

**Formulation of ODF:**

After MXDwas dissolved in a specific amount of solvent, the drug solution was added to the polymeric solution mentioned earlier. Plasticizers, such as PEG 400, sugar, and flavors were added after this stage. The medication content can be made uniform by mixing for 15 to 20 minutes in a cyclo mixer. The mixture was placed into a ready mold and let to air dry for forty-five minutes. After the film was meticulously taken out of the mold, it was inspected for flaws and trimmed to the proper size so that each strip would have the appropriate dosage (2 × 2cm2). Film samples that had cuts, air bubbles, or other flaws were not included in the investigation. HPMC E3 is used to manufacture ODF formulations MC1–MC3 at concentrations of 50–150 mg, respectively. The films made with a 100 mg concentration (MC2) were determined to have the finest formulation out of all of them. The medication release was delayed by an additional 150 mg-F4 concentration increase. HPMC E6 is used to create ODF formulations MC4–MC7 at concentrations ranging from 50–200 mg. The films made with a 100 mg concentration (MC5) were determined to be the finest formulations out of all of them. Xanthan gum is used to create ODF formulations MC8–MC11 at concentrations ranging from 50–200 mg. The films made with a 200 mg concentration (MC11) were determined to have the best formulation among them. Therefore, a formulation F12 was created that has better qualities and a more effective drug release within 15 minutes, including half the quantities of the different polymers.

**Evaluation of ODF:**

**Thickness**

A digital vernier caliper was employed to determine the thickness of the film. The thickness of the film is measured five times to ensure uniformity. The film should have a thickness of no more than 5%. The table and figure display the thickness of the rapid dissolving films for each formulation.

Table 16: Determination of Thickness, folding endurance and Surface pH of all formulations

| **F. Code** | **Thickness (µm)± SD** | **Folding endurance (folds)** | **Surface pH** |
|---|---|---|---|
| MC 1 | 98.63±1.24 | 75±5 | 6.24±0.02 |
| MC 2 | 102.54±1.36 | 83±8 | 6.31±0.01 |
| MC 3 | 103.26±1.25 | 96±10 | 6.15±0.11 |
| MC 4 | 112.58±0.39 | 102±12 | 6.31±0.12 |
| MC 5 | 124.39±0.68 | 113±7 | 6.25±0.01 |
| MC 6 | 136.48±1.45 | 125±16 | 6.11±0.05 |
| MC 7 | 98.56±1.26 | 146±14 | 6.23±0.06 |
| MC 8 | 112.43±2.14 | 159±11 | 6.03±0.03 |
| MC 9 | 136.58±1.35 | 175±10 | 6.17±0.12 |
| MC 10 | 145.32±0.46 | 162±13 | 6.25±0.01 |
| MC 11 | 120.47±0.84 | 136±15 | 6.31±0.04 |
| MC 12 | 99.68±0.57 | 125±12 | 6.05±0.05 |

Using a digital vernier caliper, the thickness of each formulation (MC1–MC12) was measured; the findings are displayed in the table. MC1–MC12 were determined to be 98.56±1.26–145.32±0.46µm thick. Based on the results of the aforementioned formulations, all of them demonstrated film thicknesses between 5 and 200 µm, which meets with the prior value's limit.

**Folding Endurance: (Kaur Mandeep etal., 2013)**

With a sharp blade, three films of each formulation were cut. A short piece of film was folded repeatedly at the same spot until it broke to assess folding durability. The value of folding endurance was determined by counting how many times the film could be folded in the same direction without breaking. Three readings were averaged and the results were computed. It was discovered that the folding endurance value of MC1–MC12 was 75±5–175±10. Based on the data gathered from the aforementioned formulations, every formulation demonstrated folding endurance values that complied with the range of 100 to 150, with the exception of MC1, MC2, MC3, MC8, MC9, and MC 10, which did not meet the previous value's limit.

**Surface pH of Films: (Birari A.E et al., 2014),**

Because changes in pH in vivo might irritate the oral mucosa, the surface pH of the films was measured to look into any potential adverse effects. An acidic or alkaline pH can do this. The film that was to be tested was put in a test tube, wet with 1.0 milliliter of distilled water, and left for thirty seconds. After contacting the formulation's surface with the pH meter's electrode and letting it acclimate for a minute, the pH was measured. For each formulation, the mean of the three findings was determined, and the standard deviation was also computed. Every film's surface pH was discovered to be between 6-7.

**Surface morphology**

The surface morphology of the MCODF sample was examined using scanning electron microscopy (SEM) at 20 kv and a magnification of X5000. The sample was made by breaking the films in liquid nitrogen, mounting it on aluminum stubs, and sputter coating it with platinum. (JEOL 5400, Tokyo, Japan).

**Tensile strength**

The greatest stress applied to the point at which the strip specimen breaks is known as its tensile strength. The formula is used to calculate it.

$$\text{Tensile strength} = \frac{\text{Load at failure} \times 100}{\text{Strip thickness} \times \text{strip width}}$$

The table displays the tensile strength of all formulations for fast dissolving films.

**Percentage elongation**

It was caluculated by

$$\text{Percentage elongation} = \frac{\text{Increase in length of strip} \times 100}{\text{Initial length of strip}}$$

The table displays the % elongation of all formulations' fast dissolving films.

Table 17: Determination of Tensile strength, Percentage elongation and In-vitro disintegration Time (sec)

| **F. Code** | **Tensile strength** | **Percentage elongation** | **In-vitro disintegration Time (sec)** |
|---|---|---|---|
| MC 1 | 45.35± | 8±1 | 25±2 |
| MC 2 | 49.85± | 9±2 | 29±4 |
| MC 3 | 52.34± | 11±3 | 19±5 |
| MC 4 | 56.41± | 12±2 | 26±3 |
| MC 5 | 57.43± | 14±5 | 21±6 |
| MC 6 | 49.83± | 13±4 | 27±2 |

| | | | |
|---|---|---|---|
| MC 7 | 53.21± | 10±2 | 23±5 |
| MC 8 | 59.86± | 14±3 | 19±4 |
| MC 9 | 61.42± | 12±1 | 24±2 |
| MC 10 | 63.27± | 11±2 | 28±3 |
| MC 11 | 57.45± | 10±3 | 29±4 |
| MC 12 | 59.84± | 13±4 | 18±2 |

**In-vitro disintegration**

The disintegration test apparatus specified in USP 2007 was used to conduct the disintegration test for each created product. The disintegration time for MC1–MC12 was 19±5–29±4 minutes. According to the data, the disintegration time increased as the polymer concentration increased. MC 3 can be used to produce MXD as a fast-dissolving delivery system based solely on the disintegration time.

**Weight variation**

A weight was determined by selecting ten videos at random and averaging them. Weighing each film, we compared its weight to the deviation's average. Table 18 shows the weight variance of all formulations' fast-dissolving films.

**Drug Content Uniformity**

The findings of the calculation of the percentage of medication content for various formulations are displayed in Table 12. With a medication content percentage of 113%, F9 was deemed the best formulation when compared to the other formulation. The formulations' 85–113% drug content percentage was displayed. based on the outcomes of the formulas mentioned above. Films' drug content should adhere to the 85–110% level specified in IP regulations (IP 2007).

**Assay**

To conduct this test, a 4 cm thin-film region was dissolved in 50 ml of pH 6.8 phosphate buffer while being stirred. Whatmann filter paper was used to filter this solution, and the filtrate was then diluted to 100 ml in a volumetric flask using the same buffer. Using a UV spectrophotometer, this solution was examined. All of the formulations' assay results are displayed in Table 19, and the values are portrayed graphically in Figure.

Table 18: Determination ofWeight variation, Drug Content Uniformity and Assay

| **F. Code** | **Weight variation** | **Drug Content Uniformity** | **Assay** |
|---|---|---|---|
| MC 1 | 59.38± | 86.95±0.26 | 97.23±0.01 |
| MC 2 | 62.43± | 92.45±0.13 | 96.45±0.05 |
| MC 3 | 65.13± | 98.76±0.21 | 98.46±0.06 |
| MC 4 | 63.02± | 100.24±0.31 | 96.85±0.08 |
| MC 5 | 59.46± | 110.46±0.27 | 99.86±0.01 |
| MC 6 | 68.75± | 104.52±0.13 | 97.03±0.04 |
| MC 7 | 66.23± | 99.76±0.42 | 96.42±0.01 |
| MC 8 | 69.43± | 98.76±0.16 | 98.01±0.02 |
| MC 9 | 70.25± | 89.82±0.34 | 97.06±0.03 |
| MC 10 | 61.23± | 112.45±0.28 | 96.43±0.04 |
| MC 11 | 65.75± | 103.24±0.31 | 97.52±0.11 |
| MC 12 | 68.43± | 105.76±0.11 | 99.73±0.12 |

**In-vitro dissolution**

The media, 900 ml of o.1 N HCL, was kept at 37 +0.5 °C with the basket spinning at 100 rpm. A 4 cm2 (2×2 cm) film sample was sliced and placed into the basket. Every two minutes, 5 milliliters of the sample were removed, and the same volume was replaced with brand-new 0.1 N HCL. After being extracted, the samples were filtered and subjected to 249 nm UV spectroscopy analysis. Data on the in vitro dissolution profiles of every formulation are shown in the table and figure. The table figure displays the percentage cumulative drug release for MC1 to MC12. The data from the marketed formulation's in-vitro dissolution profile are shown in the table and figure. The table and graphic present a comparison of the in-vitro release data between the marketed formulation and formulation MC5. Utilizing a Type I USP basket apparatus, the in vitro dissolution investigations were conducted in phosphate buffer with a 6.8 pH. In 20 minutes, MC2 with HPMC E3 released 89.76% of the drug; MC5 with HPMC E6 released 88.76%; and MC11 with Xanthan released 99.26%. Ultimately, after just 15 minutes, the formulation F12—which contained half of the optimal concentrations of HPMC E3, E6, and Xanthan gum—released 98.76% of the medication. It thus is considered to be the best formulation.

Table 19: %Cumulative Drug Release from ODF of MXD

| **Time** | **MC1** | **MC2** | **MC3** | **MC4** | **MC5** | **MC6** |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 10.23 | 12.43 | 11.43 | 8.65 | 10.54 | 8.76 |
| 4 | 26.54 | 20.61 | 21.54 | 15.48 | 20.46 | 15.43 |
| 6 | 31.05 | 28.79 | 30.61 | 23.15 | 28.76 | 22.17 |
| 8 | 39.86 | 36.48 | 40.27 | 35.47 | 36.52 | 31.04 |
| 10 | 47.85 | 45.21 | 50.43 | 42.16 | 46.18 | 39.26 |
| 12 | 59.46 | 59.78 | 59.87 | 53.27 | 52.13 | 47.51 |
| 14 | 67.21 | 69.02 | 68.75 | 61.42 | 60.13 | 58.21 |
| 16 | 75.43 | 76.42 | 79.48 | 73.21 | 72.49 | 63.12 |
| 18 | 86.73 | 84.72 | 80.31 | 80.13 | 80.45 | 75.82 |
| 20 | 91.24 | 92.43 | 91.52 | 89.76 | 98.76 | 89.61 |

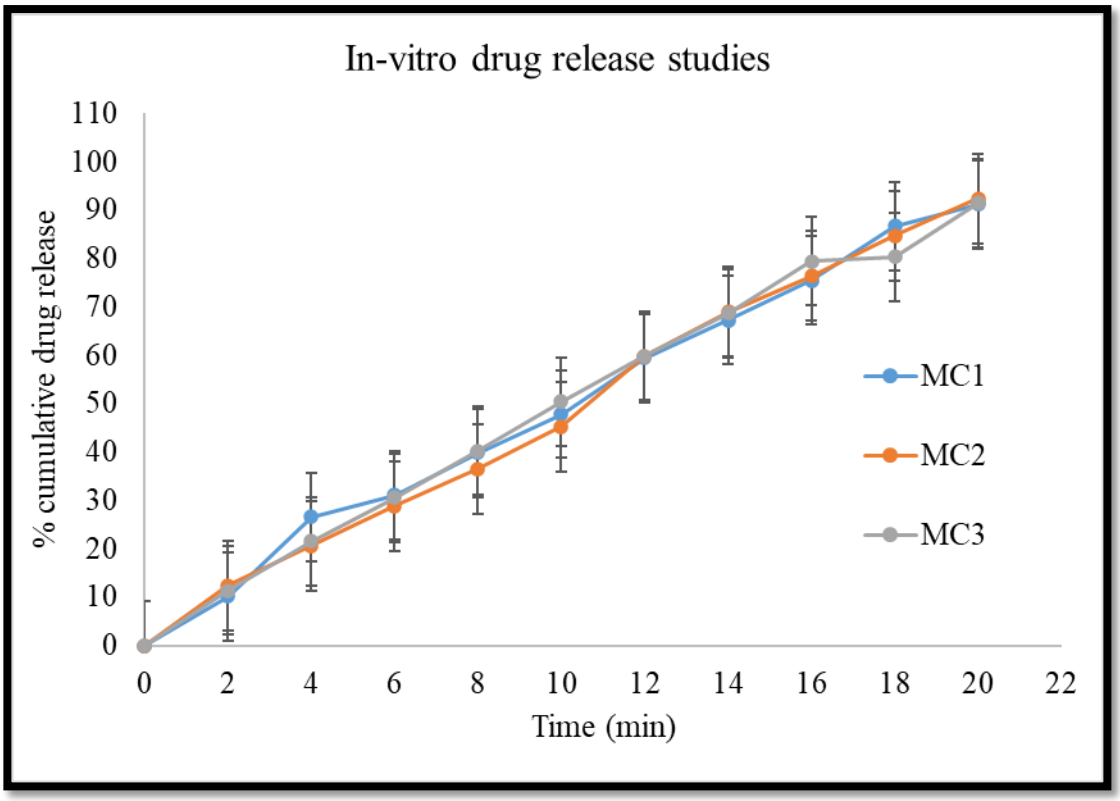


Figure 15: Invitro dissolution studies of formulations (MC1-MC3)

Table 20: %Cumulative Drug Release from ODF of MXD

| **Time** | **MC7** | **MC8** | **MC9** | **MC10** | **MC11** | **MC12** |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 9.86 | 6.89 | 8.96 | 6.53 | 10.42 | 14.23 |
| 4 | 15.43 | 12.43 | 16.75 | 12.48 | 26.35 | 28.67 |
| 6 | 26.49 | 22.51 | 26.43 | 18.97 | 37.69 | 42.31 |
| 8 | 34.02 | 31.02 | 35.97 | 26.57 | 49.75 | 68.79 |
| 10 | 42.13 | 41.27 | 47.21 | 35.46 | 63.54 | 80.43 |
| 12 | 49.86 | 50.36 | 56.31 | 48.79 | 78.95 | 98.76 |
| 14 | 60.31 | 59.76 | 67.82 | 68.75 | 86.49 | |
| 16 | 71.28 | 67.43 | 80.34 | 79.48 | 97.35 | |
| 18 | 80.36 | 80.34 | 96.57 | 89.76 | | |
| 20 | 88.76 | 89.76 | | 99.03 | | |

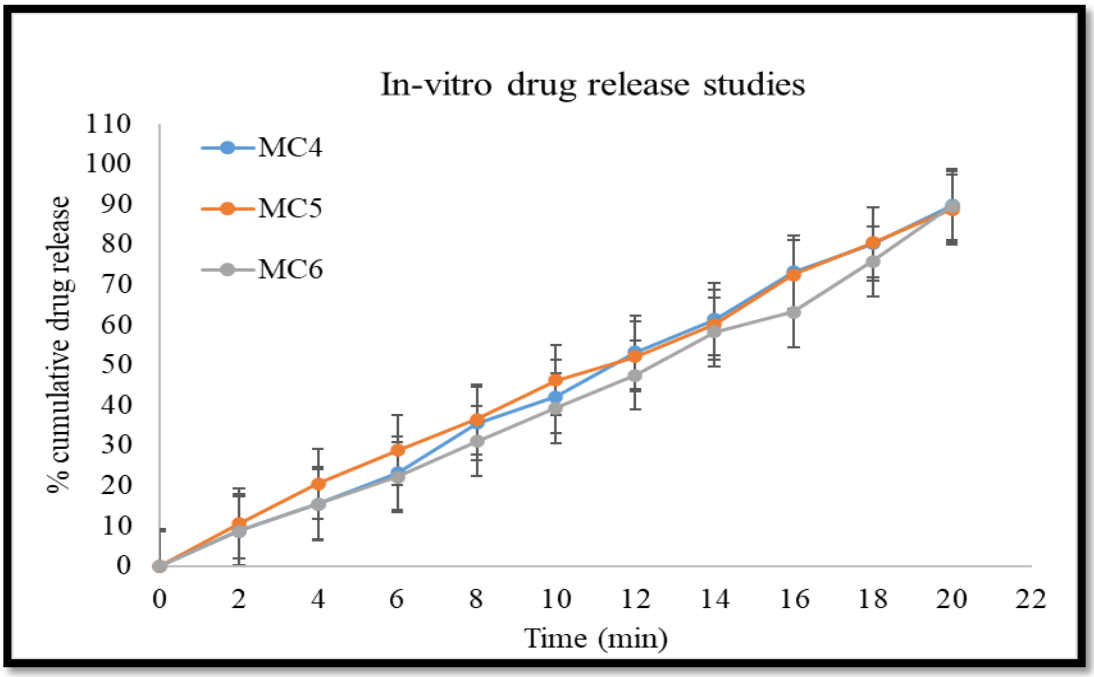


Figure 16: Invitro dissolution studies of foemulation (MC4-MC6)

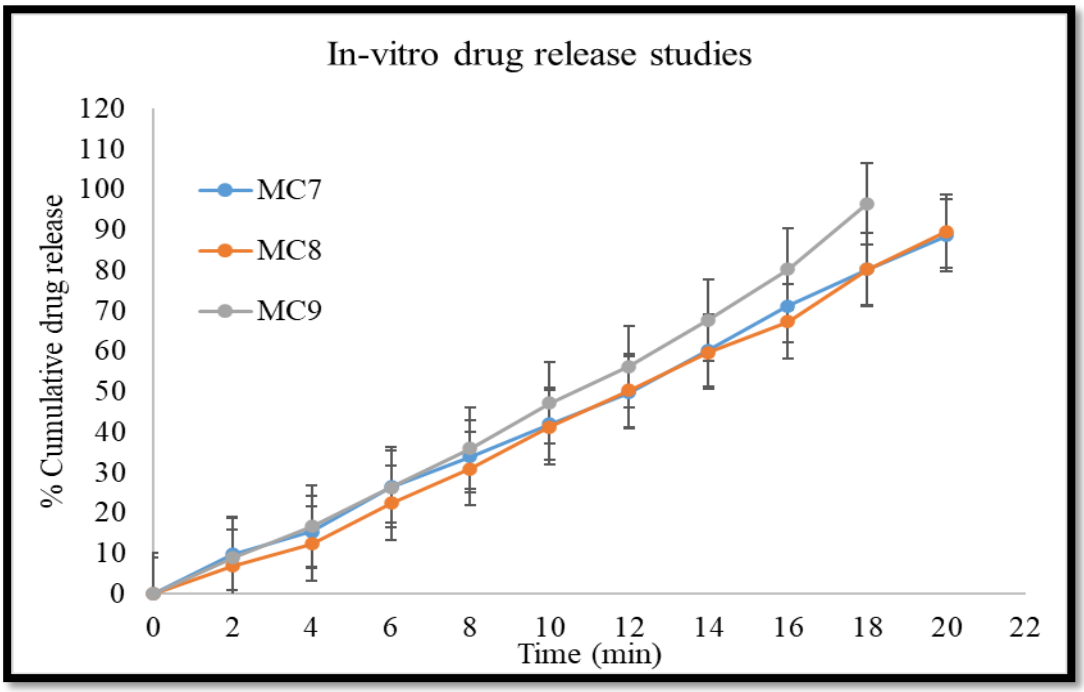


Figure 17: Invitro dissolution studies of formulations (MC7-MC9)

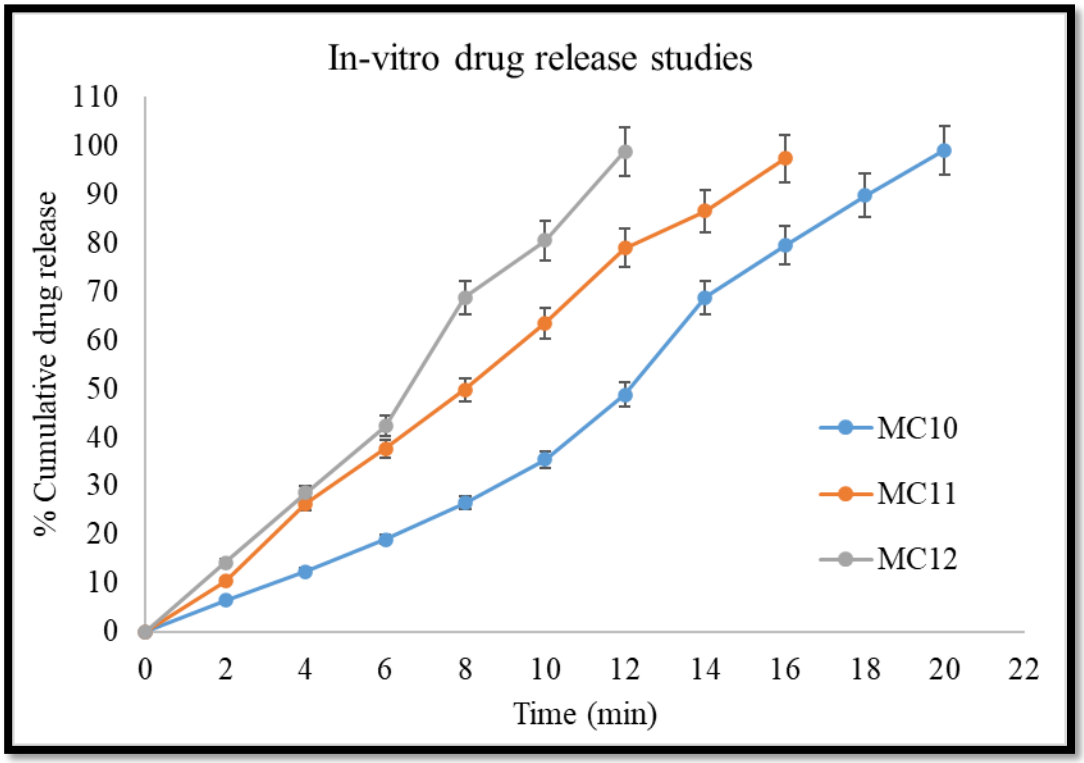


Figure 18: Invitro dissolution studies of formulations (MC10-MC12)

**SUMMARY**

Improved onset of action, reduced dosage, and improved safety and efficacy profiles of the medication are all possible with oral disintegrating films. Compared to other traditional dose forms, it is more stable, long-lasting, and rapidly dissolved. For the purpose of creating fast-dissolving films that dissolve quickly in saliva without the need for water, MXD was chosen. These films are administered sublingually. Improved patient compliance and bioavailability were the goals behind the development of MXD oral disintegrating films. With the intention of using them for elderly, bedridden, and schizophrenic patients, the formulations were created.

An antihypertensive medication is MXD. A major contributing factor to cardiovascular illnesses, such as cardiovascular stroke, is hypertension. MXD has an approximate 60–75% absolute bioavailability. After absorption, MXD, which has an average 20% bioavailability, is produced through the hydrolysis of MXD. But meals diminish the amount of MXD that is bio transformed and its bioavailability by thirty-five percent. Efforts were made to create fast-dissolving films using a natural polymer in an effort to address the issues raised above and enhance medication

solubility while minimizing negative effects.

The suitability of the medication and the carriers was determined using FTIR spectroscopy investigations. MXD rapid dissolving films could be created using the results, which indicated that there were no chemical interactions between the medication and the carriers. UV spectrophotometric techniques were used to start the development of a standard calibration curve because it is necessary for regular drug analysis. In 6.8pH phosphate buffer at 249 nm, the UV spectrophotometric technique was devised. Correlation value of 0.999 indicates that the procedure is linear in the range of 2–10 (µg/ml).

There is no drug-excipient interaction, according to the FTIR research. With the aid of solvent casting technique, water soluble polymers such as HPMC E3, HPMC E6, and Xanthum gum in different concentrations, as well as PEG 400 as a plasticizer and active ingredient, the orally disintegrating films (ODF) were created. The impact on the dissolution profile was measured using MXD. The medication disintegrated rapidly, according to the data. This work suggests that a film that dissolves when swallowed could be used to give MXD orally.

ODF formulations are smooth and easy to swallow, and their disintegration releases the entire medicine within 20 to 30 minutes. Upon disintegration, no remnant or particle debris was found. 50–150 mg of HPMC E3 was used to manufacture ODF formulations. The formulation that worked best among them was determined to be the films made with a 100 mg concentration (MC2). Drug release was postponed by a further concentration increase (150 mg–MC4). 50–200 mg ratios of HPMC E6 were used to generate ODF formulations. The 100 mg concentration film (MC5) was determined to be the most optimal formulation among them all. Xanthan gum was used in ODF formulations in dosages ranging from 50 to 200 mg. The 100 mg concentration films (MC10) were determined to be the most optimal formulations among all. This led to the development of MC 12, a formulation with improved characteristics and a more effective drug release within 15 minutes, but with half the concentrations of the component polymers.

The main goal of this work was to create a mouth-dissolving film using a combination of fundamental materials such as plasticizers, polymers, sweeteners, saliva-stimulating agents, and flavors, and morphine chloride. Soluble casting was used to create the films. It was not possible for HPMC E50 cps to give the film thickness. Good flexibility was displayed by HPMC E15.

The film's flexibility and folding endurance were not imparted by the plasticizer propylene glycol. The tensile strength, % elongation, and folding durability of PEG 400 were good. Good mechanical qualities, instantaneous drug release, folding endurance, and mouth feel were all demonstrated by the improved formulation (MC12). 99% of the medication was released in 3 minutes and the MC12 had a disintegration period of only 20 seconds, compared to an hour for the marketed version. This means that, in contrast to the traditional tablet dose form, fast drug release was accomplished for an instantaneous commencement of action.

# CONCLUSION

# CHAPTER VIII

# CONCLUSION

It was determined that the solvent casting approach was effective in producing the fast-dissolving MXD films. Orally disintegrating films (ODF) have been effectively used for patients who are immobile, schizophrenic, or traveling in areas without access to water. MXD films underwent quality control testing, including kinetic studies, stability studies, disintegration time, surface pH, folding endurance, SEM, thickness, weight fluctuation, and in-vitro diffusion. Permeation characteristics were explained by fitting the in vitro data to kinetic models. We computed and compared the coefficient of correlation for every kinetic. It was determined that it adheres to first order kinetics based on the value of the regression coefficient. After the data was further processed, Higuchi's equation revealed that the drug released via diffusion predominated with the formulation. The exponent value of the KorsmeyerPeppas model, n, describes n (value 0.45-0.85), which indicates non-Fickian diffusion. At accelerated stability conditions, the optimized formulation MC12 was found to be stable.

Improved dissolving rate and quick start of action were demonstrated by prepared films, which led to improved patient compliance, successful therapy, and increased popularity in the near future.

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## CHAPTER X

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